

Supplementary information for “The Riemann Hypothesis Manifested in Dynamical Quantum Phase Transitions”

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Contents

Notation and conventions	S1
Supplementary Note 1 Asymptotic behavior of $S_N(s)$	S2
Supplementary Note 2 Asymptotic behavior of $\mathcal{Z}(\beta, \mathcal{H}_0)$	S3
Supplementary Note 3 Entropy density $\mathcal{S}(\beta, t)$	S3
Supplementary Note 4 Construction of the preliminary oracles	S4
Supplementary Note 5 Construction of the initial state	S8
Supplementary Note 6 Construction of the evolution operator	S22
Supplementary Note 7 Order of $ \chi(s) $ for $0 < \beta < 1$	S23
Supplementary Note 8 Upper bound of $\zeta'(s)$ for $0 < \beta < 1$	S24
Supplementary References	S24

Notation and conventions

Throughout this Supplementary Information, $\log$ denotes the base-2 logarithm and $\ln$ denotes the natural logarithm. Unless otherwise specified, all variables s satisfy $\text{Re}(s) > 0$ and $\text{Re}(s) \neq 1$. We write $s = \beta + it$, where $\beta = \text{Re}(s)$ and $t = \text{Im}(s)$. The condition $\text{Re}(s) > 0$ ensures the applicability of the alternating Dirichlet series, while $\text{Re}(s) \neq 1$ excludes the zeros of the prefactor $1 - 2^{1-s}$, which all lie on the line $\text{Re}(s) = 1$. Whenever resource estimates are stated in terms of $\log |t|$, we consider the non-small t regime $|t| \geq 2$.

The Big-O notation $\mathcal{O}(\cdot)$ denotes an asymptotic upper bound, while the Big-Theta notation $\Theta(\cdot)$ denotes an asymptotically tight bound. The notation $\text{Poly}(\cdot)$ denotes a quantity bounded by a polynomial in its arguments. All precision parameters are taken in $(0, 1)$ unless stated otherwise.

The symbol $\|\cdot\|$ denotes the Euclidean norm on state vectors. The distance between two pure quantum states $|\psi_a\rangle$ and $|\psi_b\rangle$ is defined as $d(|\psi_a\rangle, |\psi_b\rangle) = \min_{\theta \in \mathbb{R}} \|\psi_a - e^{i\theta} \psi_b\| = \sqrt{2(1 - |\langle \psi_a | \psi_b \rangle|)}$, which is invariant under global phases and satisfies the triangle inequality. The distance between two unitary operators is defined as $D(U_1, U_2) = \max_{|\psi\rangle} d(U_1|\psi\rangle, U_2|\psi\rangle)$, where the maximum is taken over all normalized input states $|\psi\rangle$. A quantum state or unitary operator is prepared to precision κ if its distance from the corresponding ideal target is at most κ .

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Supplementary Note 1 – Asymptotic behavior of $S_N(s)$

This note derives the large- N scaling of $S_N(s)$, the partial sum of the alternating Dirichlet series. This asymptotic input is used below to obtain the thermodynamic entropy density $\mathcal{S}(\beta, t)$. The partial sum is

$$S_N(s) = \sum_{n=1}^N \frac{(-1)^{n+1}}{n^s}, \quad (1)$$

where $s = \beta + it$.

Case 1. At the nontrivial zeros of the zeta function, recall that $\eta(s) = (1 - 2^{1-s})\zeta(s)$. Hence at a zero of $\zeta(s)$ in the stated domain, the alternating Dirichlet series satisfies $\eta(s) = 0$, and therefore $S_N(s) = -R_N(s)$, where

$$R_N(s) = \sum_{n=N+1}^{\infty} \frac{(-1)^{n+1}}{n^s}. \quad (2)$$

Each term has the integral representation

$$\frac{1}{n^s} = \frac{1}{\Gamma(s)} \int_0^{\infty} \tau^{s-1} e^{-n\tau} d\tau. \quad (3)$$

For $\beta = \text{Re}(s) > 0$,

$$R_N(s) = \sum_{n=N+1}^{\infty} \frac{(-1)^{n+1}}{n^s} = \frac{1}{\Gamma(s)} \int_0^{\infty} \tau^{s-1} \sum_{n=N+1}^{\infty} (-1)^{n+1} e^{-n\tau} d\tau = \frac{(-1)^N}{\Gamma(s)} \int_0^{\infty} \frac{\tau^{s-1} e^{-(N+1)\tau}}{1 + e^{-\tau}} d\tau. \quad (4)$$

Using the identity $\frac{1}{1+e^{-\tau}} = \frac{1}{2} + \frac{1}{2} \tanh\left(\frac{\tau}{2}\right)$, we obtain

$$R_N(s) = \frac{(-1)^N}{\Gamma(s)} \left[\frac{1}{2} \int_0^{\infty} \tau^{s-1} e^{-(N+1)\tau} d\tau + \frac{1}{2} \int_0^{\infty} \tau^{s-1} e^{-(N+1)\tau} \tanh\left(\frac{\tau}{2}\right) d\tau \right]. \quad (5)$$

Define

$$M := \frac{1}{2} \int_0^{\infty} \tau^{s-1} e^{-(N+1)\tau} d\tau = \frac{1}{2} \Gamma(s) (N+1)^{-s}, \quad J := \frac{1}{2} \int_0^{\infty} \tau^{s-1} e^{-(N+1)\tau} \tanh\left(\frac{\tau}{2}\right) d\tau. \quad (6)$$

This leads to

$$R_N(s) = \frac{(-1)^N}{\Gamma(s)} (M + J) = (-1)^N \left(\frac{1}{2} (N+1)^{-s} + \frac{J}{\Gamma(s)} \right). \quad (7)$$

Fix $A > 0$, depending on s but not on N , and split the integral for J at $\tau = A/(N+1)$:

$$J = \frac{1}{2} \left(\int_0^{A/(N+1)} \tau^{s-1} e^{-(N+1)\tau} \tanh\left(\frac{\tau}{2}\right) d\tau + \int_{A/(N+1)}^{\infty} \tau^{s-1} e^{-(N+1)\tau} \tanh\left(\frac{\tau}{2}\right) d\tau \right) := \frac{1}{2} (I_1 + I_2). \quad (8)$$

For I_1 , use the inequality $|\tanh(\tau/2)| \leq |\tau/2|$, so

$$|I_1| \leq \int_0^{A/(N+1)} \tau^{\beta-1} e^{-(N+1)\tau} \frac{\tau}{2} d\tau = \frac{1}{2} (N+1)^{-(\beta+1)} \int_0^A u^{\beta} e^{-u} du = K_1(\beta, A) (N+1)^{-(\beta+1)}. \quad (9)$$

Here, $K_1(\beta, A) := \frac{1}{2} \int_0^A u^{\beta} e^{-u} du$ is a constant depending on A and β . For I_2 , use the trivial bound $|\tanh(\tau/2)| \leq 1$, so

$$|I_2| \leq \int_{A/(N+1)}^{\infty} \tau^{\beta-1} e^{-(N+1)\tau} d\tau = (N+1)^{-\beta} \int_A^{\infty} u^{\beta-1} e^{-u} du = K_2(\beta, A) (N+1)^{-\beta}. \quad (10)$$

Here $K_2(\beta, A) := \int_A^{\infty} u^{\beta-1} e^{-u} du$ is another constant depending on A and β . Since $\Gamma(s)$ is non-vanishing and $K_2(\beta, A) \rightarrow 0$ as $A \rightarrow \infty$, choose A large enough that $K_2(\beta, A) \leq \frac{1}{4} |\Gamma(s)|$. Thus, $|I_2| \leq \frac{1}{4} |\Gamma(s)| (N+1)^{-\beta}$. Combining the two contributions gives

$$|J| \leq \frac{1}{2} (|I_1| + |I_2|) \leq \frac{1}{2} \left(K_1(\beta, A) (N+1)^{-(\beta+1)} + \frac{1}{4} |\Gamma(s)| (N+1)^{-\beta} \right). \quad (11)$$

Consequently, for all $N \geq N_0 := \lceil 4K_1(\beta, A)/|\Gamma(s)| \rceil$,

$$|J| \leq \frac{1}{4} |\Gamma(s)| (N+1)^{-\beta}, \quad \left| \frac{J}{\Gamma(s)} \right| \leq \frac{1}{4} (N+1)^{-\beta}. \quad (12)$$

Substituting this bound into Supplementary Equation (7) gives, for $N \geq N_0$,

$$\frac{1}{4}(N+1)^{-\beta} \leq |R_N(s)| \leq \frac{3}{4}(N+1)^{-\beta}. \quad (13)$$

Therefore,

$$\lim_{N \rightarrow \infty} -\frac{1}{\log N} \ln |S_N(s)| = \lim_{N \rightarrow \infty} -\frac{1}{\log N} \ln |R_N(s)| = \beta \ln 2. \quad (14)$$

The constants used in the bounds above are for fixed s ; no uniformity in t is needed for this thermodynamic-limit statement.

Case 2. For s with $\zeta(s) \neq 0$, the partial sums converge, and in particular

$$\lim_{N \rightarrow \infty} S_N(s) = (1 - 2^{1-s}) \zeta(s) \neq 0. \quad (15)$$

It follows that

$$-\lim_{N \rightarrow \infty} \frac{1}{\log N} \ln |S_N(s)| = 0. \quad (16)$$

Thus, we obtain the final result:

$$-\lim_{N \rightarrow \infty} \frac{1}{\log N} \ln |S_N(s)| = \begin{cases} \beta \ln 2, & \text{if } \zeta(s) = 0, \\ 0, & \text{if } \zeta(s) \neq 0. \end{cases} \quad (17)$$

Supplementary Note 2 – Asymptotic behavior of $\mathcal{Z}(\beta, \mathcal{H}_0)$

This note derives the asymptotic scaling of the partition function associated with the logarithmic Hamiltonian.

Since $\mathcal{H}_0 = \sum_{n=1}^N \ln(n) |n\rangle\langle n|$, the corresponding partition function is written as

$$\mathcal{Z}(\beta, \mathcal{H}_0) = \sum_{n=1}^N n^{-\beta}. \quad (18)$$

Case 1. For $\beta > 1$, $\mathcal{Z}(\beta, \mathcal{H}_0)$ converges as

$$\lim_{N \rightarrow \infty} \mathcal{Z}(\beta, \mathcal{H}_0) = \zeta(\beta). \quad (19)$$

Thus,

$$\lim_{N \rightarrow \infty} -\frac{1}{\log N} \ln \mathcal{Z}(\beta, \mathcal{H}_0) = 0. \quad (20)$$

Case 2. For $0 < \beta < 1$, the series diverges. Specifically,

$$\int_1^{N+1} x^{-\beta} dx \leq \sum_{n=1}^N n^{-\beta} \leq 1 + \int_1^N x^{-\beta} dx, \quad (21)$$

which simplifies to

$$\frac{(N+1)^{1-\beta} - 1}{1-\beta} \leq \mathcal{Z}(\beta, \mathcal{H}_0) \leq 1 + \frac{N^{1-\beta} - 1}{1-\beta}, \quad (22)$$

yielding

$$-\lim_{N \rightarrow \infty} \frac{1}{\log N} \ln \mathcal{Z}(\beta, \mathcal{H}_0) = (\beta - 1) \ln 2. \quad (23)$$

Thus,

$$-\lim_{N \rightarrow \infty} \frac{1}{\log N} \ln \mathcal{Z}(\beta, \mathcal{H}_0) = \begin{cases} (\beta - 1) \ln 2, & 0 < \beta < 1, \\ 0, & \beta > 1. \end{cases} \quad (24)$$

Supplementary Note 3 – Entropy density $\mathcal{S}(\beta, t)$

This note combines the two asymptotic results derived in Supplementary Notes 1 and 2 to determine the free-energy density and entropy density associated with the accumulated phase factor. For $s = \beta + it$, the finite-size quantity is

$$\mathcal{F}_{1,N}(\beta, t) = -\frac{1}{\log N} \ln |\mathcal{L}_N(\beta, t)|, \quad |\mathcal{L}_N(\beta, t)| = \frac{|S_N(s)|}{\mathcal{Z}(\beta, \mathcal{H}_0)}. \quad (25)$$

Taking the thermodynamic limit gives

$$\mathcal{F}_1(\beta, t) = \begin{cases} \ln 2, & \zeta(s) = 0, \\ (1 - \beta) \ln 2, & 0 < \beta < 1, \zeta(s) \neq 0, \\ 0, & \beta > 1. \end{cases} \quad (26)$$

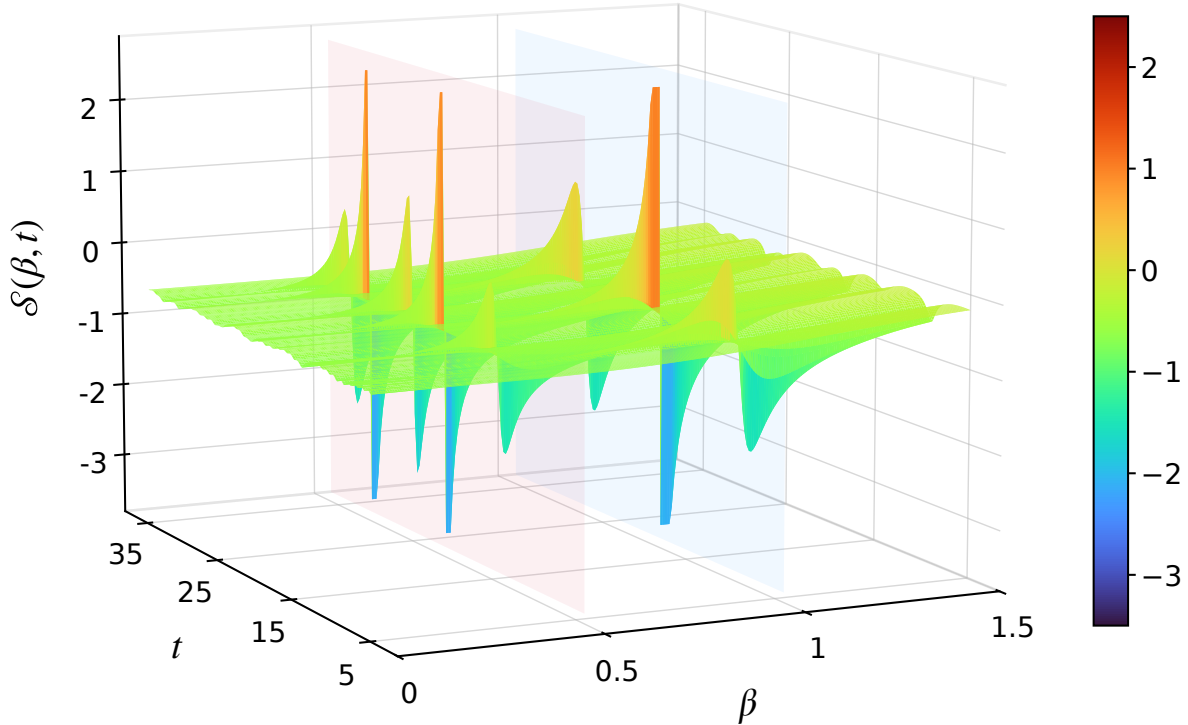
The corresponding thermodynamic entropy density is defined by

$$\mathcal{S}(\beta, t) = -\mathcal{F}_1(\beta, t) + \beta \frac{\partial \mathcal{F}_1(\beta, t)}{\partial \beta}. \quad (27)$$

Away from the nonanalytic locations, this yields

$$\mathcal{S}(\beta, t) = \begin{cases} -\ln 2, & 0 < \beta < 1, \zeta(s) \neq 0, \\ 0, & \beta > 1. \end{cases} \quad (28)$$

At the nontrivial zeros, $\mathcal{F}_1$ is nonanalytic. The smooth entropy-density formula above is therefore not assigned at the zeros themselves. In finite-size systems this nonanalytic point is rounded into a sharp local feature near the corresponding zero. Supplementary Figure 1 illustrates this finite-size entropy landscape for $N = 4096$.



Supplementary Figure 1: Entropy density $\mathcal{S}(\beta, t)$ of the quantum system as a function of t and β for $N = 4096$. The finite-size entropy surface displays rounded precursors of the discontinuous jumps associated with the nonanalyticities of $\mathcal{F}_1(\beta, t)$ at the nontrivial zeros.

Supplementary Note 4 – Construction of the preliminary oracles

This note establishes two arithmetic primitives used throughout the state-preparation and time-evolution constructions: a reversible polynomial evaluation oracle and a logarithm oracle.

Lemma 4.1 (Polynomial oracle (Methods, Lemma 1)). *Let $O(f)$ denote an oracle implementing the transformation*

$$\sum_x \alpha_x |x\rangle |0\rangle \mapsto \sum_x \alpha_x |x\rangle |f(x)\rangle, \quad (29)$$

where $f(x)$ is a polynomial of degree at most D . Suppose that the coefficients of f and the input x are specified to a_1 and a_2 bits of precision, respectively. The output is encoded with r_1 integer qubits and r_2 fractional qubits. Then $O(f)$ can be implemented using $\mathcal{O}(D^2 a_2^2 + D^2 a_1 a_2 + D(r_1 + r_2))$ gates and $\mathcal{O}(D^2 a_2 + a_1)$ ancilla qubits.

Proof. Write $f(x) = \sum_{d=0}^D c_d x^d$. The evaluation is performed by standard reversible fixed-point arithmetic. The powers of x are generated coherently in a power workspace, while a reusable product register is used to compute each contribution $c_d x^d$. After each contribution is added to the output register, the corresponding product register is uncomputed. After all contributions have been accumulated, the power-generation circuit is run in reverse to return the power workspace to its initial state.

We first estimate the cost of constructing the monomials. Multiplying an m_1 -qubit register by an m_2 -qubit register using reversible schoolbook multiplication requires $\mathcal{O}(m_1 m_2)$ gates, up to lower-order arithmetic scratch that is uncomputed within the multiplication subroutine. We store a chain of power registers

$$P_1 = x, \quad P_2 = x^2, \quad \dots, \quad P_D = x^D. \quad (30)$$

Given $P_d = x^d$, the next power is produced reversibly by computing

$$|x\rangle|P_d\rangle|0\rangle \mapsto |x\rangle|P_d\rangle|P_{d+1} = P_d x\rangle. \quad (31)$$

The register storing P_d has size at most da_2 , so this step costs $\mathcal{O}(da_2^2)$ gates. Generating the chain and then reversing the same circuit at the end gives

$$2 \sum_{d=1}^{D-1} \mathcal{O}(da_2^2) = \mathcal{O}(D^2 a_2^2). \quad (32)$$

Next, for each degree d , the monomial x^d , stored using at most da_2 qubits, is multiplied by the coefficient c_d , specified to a_1 bits of precision. This multiplication requires $\mathcal{O}(da_1 a_2)$ gates. Summing over $d = 1, \dots, D$, again including the inverse computation that uncomputes the product register after its contribution has been accumulated, gives a total gate count of

$$2 \sum_{d=1}^D \mathcal{O}(da_1 a_2) = \mathcal{O}(D^2 a_1 a_2). \quad (33)$$

The computed contribution $c_d x^d$ is then added coherently into the output register with $r_1 + r_2$ qubits. Each such reversible addition costs $\mathcal{O}(r_1 + r_2)$ gates. Since there are $D + 1$ polynomial terms, the total cost of the accumulation steps is $\mathcal{O}(D(r_1 + r_2))$. Combining the three contributions gives

$$\mathcal{O}(D^2 a_2^2 + D^2 a_1 a_2 + D(r_1 + r_2)). \quad (34)$$

For the workspace count, the power registers require

$$\sum_{d=1}^D da_2 = \mathcal{O}(D^2 a_2) \quad (35)$$

qubits. The reusable product register has size at most $Da_2 + a_1$, which is absorbed into $\mathcal{O}(D^2 a_2 + a_1)$. The reversible additions can be implemented in place on the output register. Thus the total number of ancilla qubits is

$$\mathcal{O}(D^2 a_2 + a_1). \quad (36)$$

This proves the claimed resource bounds. $\square$

Corollary 4.2 (Polynomial approximation oracle). *Let g be a real-valued function on an interval $\mathcal{I}$, and assume that $|x| \leq R$ on $\mathcal{I}$, where $R \geq 1$. Suppose that there exists a degree- D polynomial*

$$P_D(x) = \sum_{d=0}^D c_d x^d, \quad \sup_{x \in \mathcal{I}} |g(x) - P_D(x)| \leq \frac{\delta}{3}. \quad (37)$$

Let $C = \max\{1, |c_0|, \dots, |c_D|\}$. Then there is an oracle O_g acting as

$$O_g : |x\rangle|0\rangle \mapsto |x\rangle|\tilde{g}(x)\rangle, \quad |\tilde{g}(x) - g(x)| \leq \delta \quad (x \in \mathcal{I}), \quad (38)$$

where $\tilde{g}(x)$ is stored in a fixed-point output register. This oracle can be implemented using

$$\text{Poly}\left(D, a, \log C, \log R, \log \frac{1}{\delta}\right) \quad (39)$$

gates and ancilla qubits, where a denotes the total number of qubits used to encode the input x .

Proof. We evaluate a fixed-point approximation

$$\tilde{P}_D(x) = \sum_{d=0}^D \tilde{c}_d x^d, \quad |c_d - \tilde{c}_d| \leq \frac{\delta}{3C(D+1)R^D} \quad (0 \leq d \leq D). \quad (40)$$

Then, for every $x \in \mathcal{I}$,

$$\left| P_D(x) - \tilde{P}_D(x) \right| \leq \sum_{d=0}^D |c_d - \tilde{c}_d| |x|^d \leq (D+1) \max_{0 \leq d \leq D} |c_d - \tilde{c}_d| R^D \leq \frac{\delta}{3}. \quad (41)$$

Thus it is enough to use coefficient precision q , integer size r , and internal fractional precision p_{frac} satisfying

$$q = \mathcal{O}\left(\log C + \log(D+1) + D \log R + \log \frac{1}{\delta}\right), \quad r = \mathcal{O}(\log C + \log(D+1) + D \log R), \quad (42)$$

$$p_{\text{frac}} = q + Da + \left\lceil \log\left(\frac{6(D+1)}{\delta}\right) \right\rceil.$$

Thus the coefficient bit precision depends logarithmically on $1/\delta$. Here r accounts for the integer part, since all intermediate sums have magnitude at most $\mathcal{O}(C(D+1)R^D)$. The fixed-point product $\tilde{c}_d x^d$ has at most $q + Da$ fractional bits before truncation. Truncating each contribution to p_{frac} fractional bits therefore contributes at most $2^{-p_{\text{frac}}}$, and the total truncation error over all $D+1$ terms is at most

$$(D+1)2^{-p_{\text{frac}}} \leq \frac{\delta}{6}. \quad (43)$$

The final output register uses $\mathcal{O}(\log(1/\delta))$ fractional bits, so its rounding error is at most $\delta/6$. Hence

$$\left| \tilde{P}_D(x) - \tilde{g}(x) \right| \leq \frac{\delta}{3}. \quad (44)$$

Combining this with the polynomial approximation error gives

$$|g(x) - \tilde{g}(x)| \leq |g(x) - P_D(x)| + |P_D(x) - \tilde{P}_D(x)| + |\tilde{P}_D(x) - \tilde{g}(x)| \leq \delta. \quad (45)$$

Finally, applying Lemma 4.1 with coefficient precision $a_1 = q$, input precision $a_2 = a$, and accumulation size $r + p_{\text{frac}}$, then rounding to the final output register, gives

$$\begin{aligned} &\mathcal{O}(D^2 a^2 + D^2 q a + D(r + p_{\text{frac}})) \quad \text{gates,} \\ &\mathcal{O}(D^2 a + q) \quad \text{ancilla qubits.} \end{aligned} \quad (46)$$

Substituting the above bounds on q , r , and p_{frac} gives

$$\text{Poly}\left(D, a, \log C, \log R, \log \frac{1}{\delta}\right) \quad (47)$$

gates and ancilla qubits as stated. $\square$

Lemma 4.3 (Logarithm oracle (Methods, Lemma 2)). *Let L denote an oracle implementing the transformation*

$$\sum_{n=1}^N \alpha_n |n\rangle |0\rangle \mapsto \sum_{n=1}^N \alpha_n |n\rangle |\widetilde{\log(n)}\rangle, \quad (48)$$

where $\widetilde{\log(n)}$ approximates $\log(n)$ to within error η . Then L can be implemented using

$$\mathcal{O}((\log N)^3 \log^2(1/\eta) + (\log N)^2 \log^3(1/\eta)) \text{ gates and } \mathcal{O}(\log N \log^2(1/\eta)) \text{ ancilla qubits.}$$

Proof. We proceed constructively. The special cases $n = 1$ and $n = 2$ are handled separately by directly writing $\log(1) = 0$ and $\log(2) = 1$ into the output register, at negligible additional cost. Consider $3 \leq n \leq N$.

Step 1: Input partitioning. Define

$$k_1 = \left\lceil \log\left(\frac{N+1}{3}\right) \right\rceil. \quad (49)$$

For $3 \leq n < 3 \cdot 2^{k_1}$, partition the integers into k_1 subsets

$$P_\nu : \quad 3 \cdot 2^{\nu-1} \leq n_\nu < 3 \cdot 2^\nu, \quad 1 \leq \nu \leq k_1. \quad (50)$$

Since $3 \cdot 2^{k_1} \geq N + 1$, every integer $3 \leq n \leq N$ lies in exactly one partition P_ν . For $n_\nu \in P_\nu$, define

$$y = \frac{n_\nu - 2^{\nu+1}}{2^{\nu+1}}. \quad (51)$$

Then $-1/4 \leq y < 1/2$, and $n_\nu = 2^{\nu+1}(1 + y)$. Therefore,

$$\log(n_\nu) = \log(2^{\nu+1}) + \log(1 + y) = \nu + 1 + \log(1 + y). \quad (52)$$

Step 2: Taylor approximation of $\log(1 + y)$. For $|y| < 1$, the base-2 logarithm satisfies

$$\log(1 + y) = \frac{1}{\ln 2} \sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j} y^j. \quad (53)$$

Since $|y| \leq 1/2$, the truncation error after l_1 terms is bounded by

$$\left| \frac{1}{\ln 2} \sum_{j=l_1+1}^{\infty} \frac{(-1)^{j+1}}{j} y^j \right| \leq \frac{1}{\ln 2} \sum_{j=l_1+1}^{\infty} \frac{1}{j} \left(\frac{1}{2}\right)^j \leq \frac{1}{\ln 2} 2^{-l_1}. \quad (54)$$

Choosing

$$l_1 = \left\lceil \log\left(\frac{3}{\eta \ln 2}\right) \right\rceil \quad (55)$$

makes the truncation error at most $\eta/3$. Thus, for $n_\nu \in P_\nu$, we approximate $\log(n_\nu)$ by

$$\widetilde{\log(n_\nu)} = \nu + 1 + \frac{1}{\ln 2} \sum_{j=1}^{l_1} \frac{(-1)^{j+1}}{j} y^j. \quad (56)$$

Step 3: Partition flags. The partition index ν is determined by comparing n with the boundaries $3 \cdot 2^{\nu-1}$ and $3 \cdot 2^\nu$. Each comparison acts on $\mathcal{O}(k_1)$ -bit registers and can be implemented using $\mathcal{O}(k_1)$ gates and $\mathcal{O}(k_1)$ ancilla qubits, so exactly one flag is set for each input $3 \leq n \leq N$. Since there are k_1 possible partitions, constructing the one-hot partition flags requires $\mathcal{O}(k_1^2)$ gates and $\mathcal{O}(k_1)$ flag qubits.

Step 4: Controlled polynomial evaluation. For a fixed partition P_ν , Supplementary Equation (56) is a degree- l_1 polynomial in the variable y , followed by the addition of the integer offset $\nu + 1$. The dyadic number y is represented exactly using $\mathcal{O}(k_1)$ bits. The coefficients $((-1)^{j+1}/j)/\ln 2$ have magnitude bounded by a constant, and rounding each coefficient to error at most $\eta/(6l_1)$ changes the polynomial value by at most $\eta/6$ for $|y| \leq 1/2$. This requires coefficient precision $\mathcal{O}(\log(l_1/\eta)) = \mathcal{O}(\log(1/\eta))$. The output register uses $\mathcal{O}(\log k_1)$ integer bits and $\mathcal{O}(\log(1/\eta))$ fractional bits, including the integer offset; the final output rounding then contributes at most $\eta/6$. Together with the Taylor truncation error, this gives an η -approximation of $\log(n_\nu)$.

Using the resource estimate in Lemma 4.1 with $D = l_1$, $a_2 = \mathcal{O}(k_1)$, coefficient precision $a_1 = \mathcal{O}(\log(1/\eta))$, and output size $r_1 + r_2 = \mathcal{O}(\log k_1 + \log(1/\eta))$, the gate cost for one controlled polynomial evaluation is

$$\mathcal{O}(k_1^2 \log^2(1/\eta) + k_1 \log^3(1/\eta)). \quad (57)$$

We implement the oracle by scanning over $\nu = 1, \dots, k_1$. For each ν , conditioned on the corresponding partition flag, we coherently compute y , evaluate the polynomial in Supplementary Equation (56), add the offset $\nu + 1$ to the output register, and then uncompute the work registers used for this value of ν . Summing over the k_1 partitions gives

$$\mathcal{O}(k_1^3 \log^2(1/\eta) + k_1^2 \log^3(1/\eta)) \quad (58)$$

gates. The additional cost of constructing and uncomputing the partition flags, $\mathcal{O}(k_1^2)$, is dominated by this bound. The workspace for one polynomial evaluation is $\mathcal{O}(k_1 \log^2(1/\eta))$, and the partition flags require $\mathcal{O}(k_1)$ additional qubits. Since this workspace is reused across the controlled branches, the ancilla requirement is

$$\mathcal{O}(k_1 \log^2(1/\eta)). \quad (59)$$

Finally, because $k_1 = \mathcal{O}(\log N)$, the total resource bounds are

$$\mathcal{O}((\log N)^3 \log^2(1/\eta) + (\log N)^2 \log^3(1/\eta)) \text{ gates}, \quad \mathcal{O}(\log N \log^2(1/\eta)) \text{ ancilla qubits}. \quad (60)$$

This completes the construction. $\square$

Supplementary Note 5 – Construction of the initial state

This note proves the initial-state construction in two layers. First, we prepare a truncated state by Grover's recursive amplitude-splitting method, illustrated in Theorem 5.2 (Methods, Theorem 1). The splitting angles are supplied by the oracle U_m , whose implementation reduces to a partial-sum oracle for evaluation of

$$S(a, b, \beta) = \sum_{n=a}^b n^{-\beta}. \quad (61)$$

The partial-sum oracle is built from an Euler–Maclaurin approximation, endpoint-power oracles, and finite Bernoulli correction terms. Second, we combine the truncated state with the low-index component, coherently mark the desired support, and then use fixed-point amplitude amplification to obtain a deterministic preparation of the target state, yielding Theorem 5.11 (Methods, Theorem 3).

Recursive splitting

Definition 5.1. *The controlled-rotation gate $\mathcal{CR}$ implements the transformation*

$$\sum_{\theta} \alpha_{\theta} |\theta\rangle |0\rangle \mapsto \sum_{\theta} \alpha_{\theta} |\theta\rangle (\cos \theta |0\rangle + \sin \theta |1\rangle). \quad (62)$$

It can be implemented by decomposing the angle register into its binary digits and applying controlled single-qubit rotations, for example controlled R_y rotations on the target qubit.

Theorem 5.2 (Truncated state preparation (Methods, Theorem 1)). *Let $\beta > 0$, $\beta \neq 1$, and let $n_0, n_1 \in \mathbb{N}$ with $n_1 - n_0 = 2^k$ for some integer $k > 0$. Define*

$$|\psi_0\rangle_1 = \frac{1}{C_1} \sum_{n=n_0}^{n_1-1} n^{-\beta/2} |n - n_0\rangle, \quad C_1 = \sqrt{\sum_{n=n_0}^{n_1-1} n^{-\beta}}. \quad (63)$$

Then $|\psi_0\rangle_1$ can be prepared on a standard gate-based quantum computer to any precision $0 < \varepsilon < 1$, using

$$\text{Poly}\left(\log(1/\varepsilon), \log(n_1), \log\left(\frac{1}{|1-\beta|}\right), \beta, v\right), \quad (64)$$

where v denotes the number of bits of precision used to represent β . Let

$$\epsilon_{\text{sum}} = \frac{\varepsilon^2 n_1^{-\beta}}{24k^2}, \quad c_{\text{ang}} = \left\lceil \frac{1}{2} \log_{2\pi} \left(\frac{8}{\epsilon_{\text{sum}}} \right) \right\rceil = \left\lceil \frac{1}{2} \log_{2\pi} \left(\frac{192k^2 n_1^{\beta}}{\varepsilon^2} \right) \right\rceil. \quad (65)$$

The parameters are required to satisfy

$$n_0 > \lceil \beta + 2c_{\text{ang}} \rceil. \quad (66)$$

Proof. We adapt Grover's recursive amplitude-splitting method¹, replacing integration with partial summation. The 2^k computational basis states are encoded on k qubits initialized in $|0\rangle^{\otimes k}$. At iteration step m , where $0 \leq m \leq k-1$, each bit string $w \in \{0, 1\}^m$ labels the block $\{n_0 + 2^{k-m}w + i : 0 \leq i \leq 2^{k-m} - 1\}$. Define

$$c_w = \frac{1}{C_1} \sqrt{\sum_{i=0}^{2^{k-m}-1} (2^{k-m}w + i + n_0)^{-\beta}}. \quad (67)$$

The two descendants $w0$ and $w1$ correspond to the two halves of this block. Namely,

$$c_{w0} = \frac{1}{C_1} \sqrt{\sum_{i=0}^{2^{k-m-1}-1} (2^{k-m-1}(2w) + i + n_0)^{-\beta}}, \quad c_{w1} = \frac{1}{C_1} \sqrt{\sum_{i=0}^{2^{k-m-1}-1} (2^{k-m-1}(2w+1) + i + n_0)^{-\beta}}. \quad (68)$$

Therefore, $c_{w0}^2 + c_{w1}^2 = c_w^2$. For each w , we compute a rotation angle $\theta_{w,m}$ satisfying

$$\cos \theta_{w,m} = \frac{c_{w0}}{c_w}, \quad \sin \theta_{w,m} = \frac{c_{w1}}{c_w}. \quad (69)$$

Then the amplitude split

$$c_w \mapsto (c_{w0}, c_{w1}) \quad (70)$$

is implemented by a $\mathcal{CR}$ gate acting on the next qubit. The rotations for all strings $w \in \{0,1\}^m$ are performed coherently in superposition. Explicitly, the transformation at step m is

$$\begin{aligned}
& \sum_{w \in \{0,1\}^m} c_w |w\rangle |0\rangle^{\otimes(k-m)} |0\rangle \\
& \xrightarrow{U_m} \sum_w c_w |w\rangle |0\rangle^{\otimes(k-m)} |\theta_{w,m}\rangle \\
& \xrightarrow{\mathcal{CR}} \sum_w c_w |w\rangle (\cos \theta_{w,m} |0\rangle + \sin \theta_{w,m} |1\rangle) |0\rangle^{\otimes(k-m-1)} |\theta_{w,m}\rangle \\
& \xrightarrow{U_m^\dagger} \sum_w (c_{w0} |w0\rangle + c_{w1} |w1\rangle) |0\rangle^{\otimes(k-m-1)} |0\rangle.
\end{aligned} \tag{71}$$

Here, U_m denotes an oracle that computes $\theta_{w,m}$ into an angle register, which is uncomputed after applying the $\mathcal{CR}$ gate. Iterating the transformation in Supplementary Equation (71) for $m = 0, 1, \dots, k-1$ yields

$$|\psi_0\rangle_1 = \frac{1}{C_1} \sum_{n=n_0}^{n_1-1} n^{-\beta/2} |n - n_0\rangle. \tag{72}$$

Thus, the central task reduces to constructing U_m , defined as follows. The resource bound follows from the implementation of U_m in Theorem 5.4, because the k controlled rotations and their uncomputation overhead are polynomial in the same parameters. $\square$

Angle oracle

Definition 5.3 (Angle-preparation oracle U_m). *The gate U_m implements the transformation*

$$\sum_{w \in \{0,1\}^m} \alpha_w |w\rangle |0\rangle \mapsto \sum_{w \in \{0,1\}^m} \alpha_w |w\rangle |\theta_{w,m}\rangle. \tag{73}$$

For $a \leq b$, define $S(a, b, \beta) = \sum_{n=a}^b n^{-\beta}$. For each $w \in \{0,1\}^m$, let

$$S_{w0}^{(m)} = S(2^{k-m-1}(2w) + n_0, 2^{k-m-1}(2w+1) - 1 + n_0, \beta), \tag{74}$$

$$S_{w1}^{(m)} = S(2^{k-m-1}(2w+1) + n_0, 2^{k-m-1}(2w+2) - 1 + n_0, \beta), \tag{75}$$

and $S_w^{(m)} = S_{w0}^{(m)} + S_{w1}^{(m)}$. The angle $\theta_{w,m}$ is chosen so that $\theta_{w,m} \in [0, \frac{\pi}{2}]$ and

$$\tan^2(\theta_{w,m}) = \frac{S_{w1}^{(m)}}{S_{w0}^{(m)}} = \frac{S(2^{k-m-1}(2w+1) + n_0, 2^{k-m-1}(2w+2) - 1 + n_0, \beta)}{S(2^{k-m-1}(2w) + n_0, 2^{k-m-1}(2w+1) - 1 + n_0, \beta)}. \tag{76}$$

Thus,

$$\theta_{w,m} = \arctan\left(\sqrt{\frac{S_{w1}^{(m)}}{S_{w0}^{(m)}}}\right) = \arcsin\left(\sqrt{\frac{S_{w1}^{(m)}}{S_w^{(m)}}}\right). \tag{77}$$

Error analysis. In practice, U_m computes an approximation $\tilde{\theta}_{w,m}$. To bound the error introduced at one splitting step, let the ideal and approximate states after applying the controlled rotation at level m be

$$|\Phi_m\rangle = \sum_w c_w |w\rangle (\cos \theta_{w,m} |0\rangle + \sin \theta_{w,m} |1\rangle) |0\rangle^{\otimes(k-m-1)} |0\rangle, \tag{78}$$

$$|\tilde{\Phi}_m\rangle = \sum_w c_w |w\rangle (\cos \tilde{\theta}_{w,m} |0\rangle + \sin \tilde{\theta}_{w,m} |1\rangle) |0\rangle^{\otimes(k-m-1)} |0\rangle. \tag{79}$$

Then

$$\begin{aligned}
\| |\Phi_m\rangle - |\tilde{\Phi}_m\rangle \| &= \left[\sum_w c_w^2 ((\cos \theta_{w,m} - \cos \tilde{\theta}_{w,m})^2 + (\sin \theta_{w,m} - \sin \tilde{\theta}_{w,m})^2) \right]^{1/2} \\
&= \left[\sum_w c_w^2 4 \sin^2\left(\frac{\tilde{\theta}_{w,m} - \theta_{w,m}}{2}\right) \right]^{1/2} \\
&\leq \max_w |\tilde{\theta}_{w,m} - \theta_{w,m}|.
\end{aligned} \tag{80}$$

Therefore, by the triangle inequality over the k recursive splitting steps, it suffices to compute each angle $\theta_{w,m}$ to precision ε/k in order to achieve total state-preparation error at most ε . The $\mathcal{CR}$ gate is decomposed into controlled single-qubit rotations conditioned on the qubits encoding $\theta_{w,m}$. This requires $\mathcal{O}(\log(k/\varepsilon))$ gates and no additional ancilla qubits. Hence, the dominant resource cost comes from implementing U_m to precision ε/k .

Theorem 5.4 (Implementation of oracle U_m (Methods, Theorem 2)). *For $0 < \varepsilon < 1$, the gate U_m , as defined in Definition 5.3, can be implemented such that each rotation angle is estimated to within error ε/k , using*

$$\text{Poly}\left(\log(1/\varepsilon), \log(n_1), \log\left(\frac{1}{|1-\beta|}\right), \beta, v\right), \quad (81)$$

where v denotes the number of bits of precision used to represent β . The condition in Supplementary Equation (66) is assumed so that all partial-sum oracle calls used in the bisection procedure satisfy the Euler–Maclaurin remainder bound in Proposition 5.5.

Proof. The implementation uses the identity

$$\sin^2(\theta_{w,m}) = \frac{S_{w1}^{(m)}}{S_w^{(m)}}. \quad (82)$$

We compute $\theta_{w,m}$ by a coherent bisection procedure applied in superposition over all w . Namely, for each trial angle θ_g , we coherently compare θ_g with the true angle $\theta_{w,m}$ by evaluating the sign of a suitable function $d_w(\theta_g)$, and then update the bisection interval accordingly. Since $n^{-\beta}$ is decreasing for $\beta > 0$, $S_{w0}^{(m)} \geq S_{w1}^{(m)}$. Hence $\theta_{w,m} \in [0, \pi/4]$, and the bisection interval is initialized as $[0, \pi/4]$. For a trial angle θ_g , define

$$d_w(\theta_g) = \sin^2(\theta_g)S_w^{(m)} - S_{w1}^{(m)}. \quad (83)$$

The sign of $d_w(\theta_g)$ determines whether θ_g is smaller or larger than $\theta_{w,m}$. Repeating this coherent bisection procedure for $\mathcal{O}(\log(k/\varepsilon))$ steps determines $\theta_{w,m}$ to precision ε/k , provided that each comparison is evaluated with sufficient precision.

To determine the required comparison precision, set $\delta = \varepsilon/k$. For any $\theta, \theta' \in [0, \pi/4]$ with $|\theta - \theta'| \geq \delta$, assume without loss of generality that $\theta' \geq \theta$. Then

$$|\sin^2 \theta' - \sin^2 \theta| = \sin(\theta + \theta') \sin(\theta' - \theta) \geq \sin^2(\theta' - \theta) \geq \frac{\delta^2}{2}, \quad (84)$$

where we used $\sin x \geq x/\sqrt{2}$ for $x \in [0, \pi/4]$. Moreover, $S_w^{(m)} \geq n_1^{-\beta}$, because every block contains at least one integer $n \leq n_1 - 1$. Therefore, whenever $|\theta_g - \theta_{w,m}| \geq \varepsilon/k$, the ideal value of $d_w(\theta_g)$ has magnitude at least

$$\frac{\varepsilon^2}{2k^2} n_1^{-\beta}. \quad (85)$$

It is therefore sufficient to evaluate $d_w(\theta_g)$ with error smaller than this separation. If $|\theta_g - \theta_{w,m}| < \varepsilon/k$, then the current trial angle is already within the target accuracy, so an unstable sign decision in this near-degenerate case does not affect the final accuracy guarantee.

Since $0 \leq \sin^2(\theta_g) \leq 1$ and $S_w^{(m)} \leq n_1$, the following two precision requirements are sufficient:

- (i) compute each partial sum S to precision $\epsilon_{\text{sum}} = \frac{\varepsilon^2 n_1^{-\beta}}{24k^2}$, (ii) compute $\sin^2(\theta_g)$ to precision $\epsilon_{\text{sin}} = \frac{\varepsilon^2 n_1^{-1-\beta}}{24k^2}$.

Indeed, we evaluate $S_w^{(m)}$ directly as one partial sum over the whole block, and evaluate $S_{w1}^{(m)}$ as a second partial sum. The product term in Supplementary Equation (83) is controlled as follows. Let $q = \sin^2(\theta_g)$, and let $\tilde{q}$ and $\tilde{S}_w$ be the approximations. Then

$$|\tilde{q}\tilde{S}_w - qS_w| \leq |q| |\tilde{S}_w - S_w| + |\tilde{S}_w| |\tilde{q} - q|. \quad (86)$$

Since $0 \leq q \leq 1$ and $|\tilde{S}_w| \leq n_1 + \epsilon_{\text{sum}}$, this gives

$$|\tilde{q}\tilde{S}_w - qS_w| \leq \epsilon_{\text{sum}} + (n_1 + \epsilon_{\text{sum}})\epsilon_{\text{sin}}. \quad (87)$$

Including the approximation error in $S_{w1}^{(m)}$, the total comparison error is bounded by

$$2\epsilon_{\text{sum}} + (n_1 + \epsilon_{\text{sum}})\epsilon_{\text{sin}} \leq 2\epsilon_{\text{sum}} + 2n_1\epsilon_{\text{sin}} = \frac{\varepsilon^2 n_1^{-\beta}}{6k^2}, \quad (88)$$

where the inequality uses $\epsilon_{\text{sum}} \leq n_1$, which follows from $0 < \varepsilon < 1$, $\beta > 0$, and $k, n_1 \geq 1$. This is smaller than the separation above.

Step 1: Calculation of $\sin^2(\theta_g)$. Requirement (ii) is achieved by approximating $\sin^2(\theta)$ on $0 \leq \theta \leq \pi/4$. Set $\epsilon_{\sin} := \varepsilon^2 n_1^{-1-\beta} / (24k^2)$. We use the Taylor expansion

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2} = \sum_{j=1}^{\infty} d_j \theta^{2j}, \quad d_j = \frac{(-1)^{j+1} 2^{2j}}{2(2j)!}. \quad (89)$$

Let $P_{l_2}(\theta) = \sum_{j=1}^{l_2} d_j \theta^{2j}$. Since $0 \leq \theta \leq \pi/4$, the truncation error satisfies

$$|\sin^2(\theta) - P_{l_2}(\theta)| \leq \frac{1}{2} \sum_{j=l_2+1}^{\infty} \frac{(\pi/2)^{2j}}{(2j)!}. \quad (90)$$

Thus choosing

$$l_2 = \mathcal{O}(\log(k/\varepsilon) + (1 + \beta) \log n_1) = \mathcal{O}(\log(1/\epsilon_{\sin})) \quad (91)$$

makes the truncation error at most $\epsilon_{\sin}/3$.

The polynomial P_{l_2} has degree $2l_2$ on an interval contained in $[-1, 1]$, and its coefficients are bounded by a constant. By Corollary 4.2, applied with target precision $\delta = \epsilon_{\sin}$, $D = 2l_2$, $R = 1$, $C = \mathcal{O}(1)$, and input precision

$$a = \mathcal{O}(\log(1/\epsilon_{\sin})), \quad (92)$$

the oracle for $\sin^2(\theta_g)$ uses

$$\text{Poly}(\log(1/\epsilon_{\sin})) = \text{Poly}(\log(1/\varepsilon), \log n_1, \beta) \quad (93)$$

gates and ancilla qubits. This gives the precision required in (ii).

Step 2: Calculation of the partial sums. Requirement (i) is achieved by the partial-sum oracle $\mathcal{S}(\beta)$ in Lemma 5.9. In this application,

$$\epsilon_{\text{sum}} = \frac{\varepsilon^2 n_1^{-\beta}}{24k^2}, \quad l_3 = \left\lceil \frac{1}{2} \log_{2\pi} \left(\frac{8}{\epsilon_{\text{sum}}} \right) \right\rceil = c_{\text{ang}}. \quad (94)$$

Thus the assumption $n_0 > \lceil \beta + 2c_{\text{ang}} \rceil$ guarantees that every partial-sum query in the bisection procedure satisfies the Euler–Maclaurin remainder condition in Proposition 5.5.

By Lemma 5.9, each call to $\mathcal{S}(\beta)$ with precision ϵ_{sum} uses

$$\text{Poly} \left(\log(1/\epsilon_{\text{sum}}), \log n_1, \log \left(\frac{1}{|1 - \beta|} \right), \beta, v \right) \quad (95)$$

gates and ancilla qubits. Since

$$\log(1/\epsilon_{\text{sum}}) = \mathcal{O}(\log(k/\varepsilon) + \beta \log n_1), \quad (96)$$

this cost is bounded by

$$\text{Poly} \left(\log(1/\varepsilon), \log n_1, \log \left(\frac{1}{|1 - \beta|} \right), \beta, v \right). \quad (97)$$

It remains to combine the costs in the coherent bisection. Each bisection step uses two calls to the partial-sum oracle $\mathcal{S}(\beta)$, one evaluation of $\sin^2(\theta_g)$, and reversible fixed-point arithmetic to form and compare $d_w(\theta_g)$. The number of bisection steps is $\mathcal{O}(\log(k/\varepsilon))$. Hence the full coherent bisection procedure has total gate complexity polynomial in

$$\log(1/\varepsilon), \quad \log n_1, \quad \log \left(\frac{1}{|1 - \beta|} \right), \quad \beta, \quad v. \quad (98)$$

All work registers are uncomputed after each comparison, so the ancilla count remains polynomial in the same parameters. This proves the claimed implementation of U_m . $\square$

Partial-sum oracle

The following results provide the arithmetic ingredients used by the partial-sum oracle $\mathcal{S}(\beta)$. First, Proposition 5.5 derives an Euler–Maclaurin approximation for $S(a, b, \beta)$, with the required Bernoulli-number estimates summarized in Corollary 5.6. Second, Proposition 5.7 constructs the endpoint-power oracles needed to evaluate the boundary terms in this approximation. Third, Proposition 5.8 evaluates the finite correction sum. These ingredients are then assembled in Lemma 5.9, where the partial-sum oracle used above is formalized.

Proposition 5.5 (Euler–Maclaurin approximation of S). *For $\beta > 0$, $\beta \neq 1$, $0 < \epsilon < 1$, and integers $a \leq b$, define*

$$S(a, b, \beta) = \sum_{n=a}^b n^{-\beta}. \quad (99)$$

There exists an approximation $\tilde{S}(a, b, \beta)$ satisfying

$$\left| S(a, b, \beta) - \tilde{S}(a, b, \beta) \right| \leq \frac{\epsilon}{2}, \quad (100)$$

where

$$\tilde{S}(a, b, \beta) = \frac{b^{1-\beta} - a^{1-\beta}}{1-\beta} + \frac{a^{-\beta} + b^{-\beta}}{2} - \sum_{r=1}^{l_3} \frac{B_{2r} \Gamma(\beta + 2r - 1)}{(2r)! \Gamma(\beta)} (b^{-\beta-2r+1} - a^{-\beta-2r+1}), \quad (101)$$

with

$$l_3 = \left\lceil \frac{1}{2} \log_{2\pi} \left(\frac{8}{\epsilon} \right) \right\rceil, \quad (102)$$

provided $a > \lceil \beta + 2l_3 \rceil$.

Proof. Recall the Euler–Maclaurin summation formula:

$$\sum_{n=a}^b f(n) = \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} + \sum_{r=1}^{l_3} \frac{B_{2r}}{(2r)!} (f^{(2r-1)}(b) - f^{(2r-1)}(a)) + R_{l_3}, \quad (103)$$

where

$$R_{l_3} = \frac{(-1)^{l_3+1}}{(2l_3)!} \int_a^b B_{2l_3}(\{x\}) f^{(2l_3)}(x) dx. \quad (104)$$

Here $B_{2l_3}(\{x\})$ denotes the periodic Bernoulli polynomial and $\{x\}$ denotes the fractional part of x . Set $f(x) = x^{-\beta}$. Since

$$f^{(q)}(x) = (-1)^q \frac{\Gamma(\beta + q)}{\Gamma(\beta)} x^{-\beta-q}, \quad (105)$$

applying the Euler–Maclaurin formula gives

$$S(a, b, \beta) = \tilde{S}(a, b, \beta) + R_{l_3}, \quad (106)$$

where $\tilde{S}(a, b, \beta)$ is the truncated Euler–Maclaurin approximation defined in Supplementary Equation (101), and the remainder is

$$R_{l_3} = -\frac{1}{(2l_3)!} \int_a^b B_{2l_3}(\{x\}) \frac{d^{2l_3}}{dx^{2l_3}} x^{-\beta} dx. \quad (107)$$

We bound this remainder. Since

$$\left| \frac{d^{2l_3}}{dx^{2l_3}} x^{-\beta} \right| = \frac{\Gamma(\beta + 2l_3)}{\Gamma(\beta)} x^{-\beta-2l_3}, \quad (108)$$

and

$$\sup_{0 \leq y \leq 1} |B_{2l_3}(y)| = |B_{2l_3}|, \quad (109)$$

we have

$$|R_{l_3}| \leq \frac{|B_{2l_3}|}{(2l_3)!} \frac{\Gamma(\beta + 2l_3)}{\Gamma(\beta)} \int_a^b x^{-\beta-2l_3} dx. \quad (110)$$

Using

$$|B_{2l_3}| = \frac{2(2l_3)!}{(2\pi)^{2l_3}} \zeta(2l_3) < \frac{4(2l_3)!}{(2\pi)^{2l_3}}, \quad (111)$$

and

$$\int_a^b x^{-\beta-2l_3} dx \leq \frac{a^{-\beta-2l_3+1}}{\beta + 2l_3 - 1}, \quad (112)$$

we obtain

$$|R_{l_3}| \leq \frac{4}{(2\pi)^{2l_3}} \frac{\Gamma(\beta + 2l_3)}{(\beta + 2l_3 - 1)\Gamma(\beta)} a^{-\beta-2l_3+1} = \frac{4}{(2\pi)^{2l_3}} \frac{\beta(\beta + 1) \cdots (\beta + 2l_3 - 2)}{a^{\beta+2l_3-1}}. \quad (113)$$

If $a > \beta + 2l_3$, then each factor $\beta, \beta + 1, \dots, \beta + 2l_3 - 2$ is smaller than a . Hence

$$\frac{\beta(\beta + 1) \cdots (\beta + 2l_3 - 2)}{a^{\beta+2l_3-1}} < \frac{a^{2l_3-1}}{a^{\beta+2l_3-1}} = a^{-\beta} < 1. \quad (114)$$

Therefore

$$|R_{l_3}| \leq \frac{4}{(2\pi)^{2l_3}}. \quad (115)$$

With l_3 chosen as in Supplementary Equation (102), this bound is at most $\epsilon/2$. Consequently,

$$|S(a, b, \beta) - \tilde{S}(a, b, \beta)| = |R_{l_3}| \leq \epsilon/2, \quad (116)$$

which proves the claim. $\square$

Corollary 5.6. *The Bernoulli numbers satisfy $B_0 = 1$, and for all $q \geq 1$,*

$$\sum_{p=0}^q \binom{q+1}{p} B_p = 0. \quad (117)$$

Moreover, the Bernoulli numbers $B_0, B_1, \dots, B_p$ can be computed classically using the Akiyama–Tanigawa algorithm² with $\mathcal{O}(p^2)$ arithmetic operations, and with bit complexity polynomial in p and in the requested output precision.

Proposition 5.7. *For $0 < \epsilon < 1$, the endpoint-power oracle*

$$\sum_{x=n_0}^{n_1-1} \alpha_x |x\rangle |0\rangle \mapsto \sum_{x=n_0}^{n_1-1} \alpha_x |x\rangle |\widetilde{x^{-\beta-2l_3+1}}\rangle \quad (118)$$

can be implemented with pointwise error

$$\left| \widetilde{x^{-\beta-2l_3+1}} - x^{-\beta-2l_3+1} \right| < \frac{\epsilon|1-\beta|}{16x^{2l_3}}, \quad n_0 \leq x \leq n_1 - 1, \quad (119)$$

using

$$\text{Poly}\left(\log(n_1), \log\left(\frac{1}{|1-\beta|}\right), \log(1/\epsilon), \beta, v\right) \quad (120)$$

gates and ancilla qubits.

Proof. Let

$$\lambda = \beta + 2l_3 - 1. \quad (121)$$

Since $l_3 \geq 1$, we have $\lambda > 0$. For $n_0 \leq x \leq n_1 - 1$,

$$x^{-\beta-2l_3+1} = x^{-\lambda} = 2^{-\lambda \log x}, \quad (122)$$

where $\log$ denotes the base-2 logarithm. The oracle is therefore implemented by computing $\log x$, multiplying the result by $-\lambda$, evaluating 2^z at the resulting exponent, and then uncomputing the intermediate registers.

Use one uniform precision for the whole input range:

$$\Delta = \frac{\epsilon|1-\beta|}{16n_1^{2l_3}}. \quad (123)$$

Since $x \leq n_1$,

$$\Delta \leq \frac{\epsilon|1-\beta|}{16x^{2l_3}}. \quad (124)$$

Thus it suffices to approximate $x^{-\lambda}$ to additive error less than Δ . If $\Delta \geq 1$, even the output 0 is within error Δ , since $0 < x^{-\lambda} \leq 1$. Thus we consider the nontrivial case $\Delta < 1$ below.

First compute $\log x$ to precision

$$\frac{\Delta}{16\lambda}. \quad (125)$$

After multiplication by $-\lambda$, this contributes at most $\Delta/16$ error to the exponent. We perform the fixed-point multiplication by $-\lambda$ with rounding error at most another $\Delta/16$. Hence the computed exponent $\tilde{z}$ satisfies

$$|\tilde{z} + \lambda \log x| \leq \frac{\Delta}{8}. \quad (126)$$

Since $\Delta < 1$, Supplementary Equation (126) implies that the computed exponent $\tilde{z}$ lies in the unit enlargement of the exact exponent range:

$$-\lambda \log n_1 \leq -\lambda \log x \leq 0, \quad -\lambda \log n_1 - 1 \leq \tilde{z} \leq 1. \quad (127)$$

We approximate 2^z on the latter interval. Set $M = \lambda \log n_1 + 1$, so this interval is contained in $[-M, M]$. We use the explicit Taylor polynomial

$$P_L(z) = \sum_{j=0}^L \frac{(\ln 2)^j}{j!} z^j, \quad (128)$$

with

$$L = \left\lceil 8 \left(M + \log \left(\frac{8}{\Delta} \right) \right) \right\rceil. \quad (129)$$

For $|z| \leq M$, write $2^z = \exp((\ln 2)z)$. Taylor's theorem with Lagrange remainder gives

$$2^z = P_L(z) + \frac{\exp(\rho)}{(L+1)!} ((\ln 2)z)^{L+1}, \quad (130)$$

where ρ lies between 0 and $(\ln 2)z$. Therefore

$$|2^z - P_L(z)| \leq 2^M \frac{((\ln 2)M)^{L+1}}{(L+1)!}. \quad (131)$$

Using $(L+1)! \geq ((L+1)/e)^{L+1}$, and noting that this choice gives $(e \ln 2)M/(L+1) \leq 1/4$, we obtain

$$|2^z - P_L(z)| \leq 2^M 4^{-(L+1)} \leq \frac{\Delta}{6}. \quad (132)$$

The polynomial P_L has degree

$$L = \text{Poly}(M, \log(1/\Delta)), \quad (133)$$

and its coefficients are bounded by a constant. Since $\Delta/6 = (\Delta/2)/3$, the truncation error satisfies the hypothesis of Corollary 4.2 with target precision $\delta = \Delta/2$. Applying the corollary on the interval $[-M, M]$ (so $R = M$) gives an oracle evaluating 2^z to precision $\Delta/2$. The required resources are polynomial in L , the input precision for z , $\log M$, and $\log(1/\Delta)$.

The exponent error is propagated on the interval $[-\lambda \log n_1 - 1, 1]$,

$$\left| \frac{d}{dz} 2^z \right| = (\ln 2) 2^z \leq 2 \ln 2 < 2. \quad (134)$$

Therefore, by the mean-value theorem and Supplementary Equation (126),

$$\left| 2^{\tilde{z}} - 2^{-\lambda \log x} \right| \leq 2 |\tilde{z} + \lambda \log x| \leq \frac{\Delta}{4}. \quad (135)$$

Combining this with the $\Delta/2$ -accurate polynomial evaluation gives

$$\left| \widetilde{x^{-\lambda}} - x^{-\lambda} \right| < \Delta. \quad (136)$$

By Supplementary Equation (123), this implies

$$\left| \widetilde{x^{-\beta-2l_3+1}} - x^{-\beta-2l_3+1} \right| < \frac{\epsilon |1-\beta|}{16x^{2l_3}}, \quad (137)$$

which proves the claimed precision.

For the resource scaling, Supplementary Equation (123) gives

$$\log(1/\Delta) = \mathcal{O} \left(\log(1/\epsilon) + \log \left(\frac{1}{|1-\beta|} \right) + l_3 \log n_1 \right). \quad (138)$$

Since $l_3 = \mathcal{O}(\log(1/\epsilon))$, this is polynomial in

$$\log(n_1), \quad \log \left(\frac{1}{|1-\beta|} \right), \quad \log(1/\epsilon), \quad \beta. \quad (139)$$

Moreover,

$$\log \left(\frac{16\lambda}{\Delta} \right) = \mathcal{O}(\log(1/\Delta) + \log \lambda), \quad (140)$$

which is polynomial in the same parameters. The fixed-point arithmetic involving $\lambda = \beta + 2l_3 - 1$ uses the v -bit description of β , which is the source of the v -dependence below. The degree L in Supplementary Equation (129) is polynomial in

$$\lambda \log n_1 \quad \text{and} \quad \log(1/\Delta). \quad (141)$$

Thus the logarithm oracle, the fixed-point multiplication by $-\lambda$, and the polynomial evaluation of 2^z all use

$$\text{Poly}\left(\log(n_1), \log\left(\frac{1}{|1-\beta|}\right), \log(1/\epsilon), \beta, v\right) \quad (142)$$

gates and ancilla qubits. Finally, all intermediate registers are uncomputed after the output is written, so the oracle is implemented cleanly. This proves the proposition. $\square$

Proposition 5.8. For $0 < \epsilon < 1$ and $n_0 > \lceil \beta + 2l_3 \rceil$, define

$$P_b(\beta) = \frac{1}{1-\beta} b^{2l_3} + \frac{1}{2} b^{2l_3-1} - \sum_{r=1}^{l_3} C_r(\beta) b^{2l_3-2r}, \quad (143)$$

$$P_a(\beta) = \frac{1}{1-\beta} a^{2l_3} - \frac{1}{2} a^{2l_3-1} - \sum_{r=1}^{l_3} C_r(\beta) a^{2l_3-2r}, \quad (144)$$

where

$$C_r(\beta) = \frac{B_{2r} \Gamma(\beta + 2r - 1)}{(2r)! \Gamma(\beta)}. \quad (145)$$

The oracles

$$\sum_{b=n_0}^{n_1-1} \alpha_b |b\rangle |0\rangle \mapsto \sum_{b=n_0}^{n_1-1} \alpha_b |b\rangle |P_b(\beta)\rangle \quad (146)$$

and

$$\sum_{a=n_0}^{n_1-1} \alpha_a |a\rangle |0\rangle \mapsto \sum_{a=n_0}^{n_1-1} \alpha_a |a\rangle |P_a(\beta)\rangle \quad (147)$$

can be implemented to precision $\epsilon/8$ using

$$\text{Poly}\left(\log(1/\epsilon), \log(n_1), \log\left(\frac{1}{|1-\beta|}\right), \beta, v\right) \quad (148)$$

gates and ancilla qubits, where v denotes the number of bits used to represent β .

Proof. We describe the construction for $P_b(\beta)$; the construction for $P_a(\beta)$ is identical. For fixed l_3 , $P_b(\beta)$ is a degree- $2l_3$ polynomial in b . Its coefficients are

$$\frac{1}{1-\beta}, \quad \frac{1}{2}, \quad C_r(\beta) = \frac{B_{2r}}{(2r)!} \prod_{j=0}^{2r-2} (\beta + j), \quad 1 \leq r \leq l_3. \quad (149)$$

By Corollary 5.6, the Bernoulli numbers up to B_{2l_3} can be computed classically in time polynomial in l_3 . The products $\prod_{j=0}^{2r-2} (\beta + j)$ and the factor $1/(1-\beta)$ can be precomputed from the v -bit description of β to the required precision with cost polynomial in

$$\log(1/\epsilon), \quad \log\left(\frac{1}{|1-\beta|}\right), \quad \beta, \quad v. \quad (150)$$

We first record a uniform range bound that will also be used below. Under the condition $b \geq n_0 > \beta + 2l_3$, the Bernoulli correction terms in $P_b(\beta)$ are dominated by a geometric series. Define

$$T_b(r) = \frac{B_{2r} \Gamma(\beta + 2r - 1)}{(2r)! \Gamma(\beta)} b^{2l_3-2r}, \quad 1 \leq r \leq l_3. \quad (151)$$

For $1 \leq r < l_3$, the condition $b > \beta + 2l_3$ implies both $(\beta + 2r - 1)/b < 1$ and $(\beta + 2r)/b < 1$, which will be used in the ratio estimate below. Then, for $1 \leq r < l_3$,

$$\left| \frac{T_b(r+1)}{T_b(r)} \right| = \left| \frac{B_{2r+2}}{B_{2r}} \right| \frac{(\beta + 2r - 1)(\beta + 2r)}{(2r+2)(2r+1)} \frac{1}{b^2} < \frac{(\beta + 2r - 1)(\beta + 2r)}{(2\pi)^2 b^2} \frac{1 - 2^{-2r}}{1 - 2^{1-2(r+1)}} < \frac{1}{2\pi^2}, \quad (152)$$

where we used the standard Bernoulli-number estimate

$$|B_{2r}| = \frac{2(2r)!}{(2\pi)^{2r}} \zeta(2r) \quad (153)$$

together with $b > \beta + 2l_3$. Hence

$$\left| \sum_{r=1}^{l_3} \frac{B_{2r} \Gamma(\beta + 2r - 1)}{(2r)! \Gamma(\beta)} b^{2l_3 - 2r} \right| \leq \frac{|T_b(1)|}{1 - \frac{1}{2\pi^2}} = \frac{\beta b^{2l_3 - 2}}{12 \left(1 - \frac{1}{2\pi^2}\right)} < \frac{b^{2l_3 - 1}}{2}. \quad (154)$$

Since $b > |1 - \beta|$, this implies

$$|P_b(\beta)| < \frac{2b^{2l_3}}{|1 - \beta|} \leq \frac{2n_1^{2l_3}}{|1 - \beta|}. \quad (155)$$

The same argument with b replaced by a gives

$$|P_a(\beta)| < \frac{2a^{2l_3}}{|1 - \beta|}. \quad (156)$$

These finite polynomials can be evaluated reversibly. The coefficient magnitudes have logarithms bounded by a polynomial in

$$\log(1/\epsilon), \quad \log\left(\frac{1}{|1 - \beta|}\right), \quad \beta \quad (157)$$

because $l_3 = \mathcal{O}(\log(1/\epsilon))$, the Bernoulli numbers are computed classically by Corollary 5.6, and the remaining coefficient factors are products of terms $\beta + j$ with $0 \leq j \leq 2l_3 - 2$. Thus the coefficient bound C for these polynomials has $\log C$ polynomial in the displayed parameters without introducing an additional v -dependence. Producing the required fixed-point coefficient approximations from the v -bit description of β , however, has cost polynomial also in v . Since $P_b(\beta)$ and $P_a(\beta)$ are already finite polynomials, their polynomial-approximation error is zero. By Corollary 4.2, applied with target precision $\delta = \epsilon/8$ and

$$D = 2l_3, \quad R = n_1, \quad a = \mathcal{O}(\log n_1), \quad (158)$$

both $P_b(\beta)$ and $P_a(\beta)$ can be evaluated using

$$\text{Poly}\left(\log(1/\epsilon), \log(n_1), \log\left(\frac{1}{|1 - \beta|}\right), \beta, v\right) \quad (159)$$

gates and ancilla qubits. This proves the claim. $\square$

Lemma 5.9 (Partial-sum oracle). *Let $\mathcal{S}(\beta)$ be an oracle implementing*

$$\sum_{n_0 \leq a \leq b \leq n_1 - 1} \alpha_{ab} |a\rangle |b\rangle |0\rangle \mapsto \sum_{n_0 \leq a \leq b \leq n_1 - 1} \alpha_{ab} |a\rangle |b\rangle |\hat{S}(a, b, \beta)\rangle. \quad (160)$$

Assume $0 < \epsilon < 1$ and $n_0 > \lceil \beta + 2l_3 \rceil$, with l_3 defined in Supplementary Equation (102). Then $\mathcal{S}(\beta)$ can be implemented so that $\hat{S}(a, b, \beta)$ is an ϵ -approximation of $S(a, b, \beta)$, using

$$\text{Poly}\left(\log(1/\epsilon), \log(n_1), \log\left(\frac{1}{|1 - \beta|}\right), \beta, v\right) \quad (161)$$

gates and ancilla qubits.

Proof. Let $\lambda = \beta + 2l_3 - 1$. Supplementary Equation (101) can be factorized as

$$\tilde{S}(a, b, \beta) = b^{-\lambda} P_b(\beta) - a^{-\lambda} P_a(\beta), \quad (162)$$

with $P_b(\beta)$ and $P_a(\beta)$ defined in Supplementary Equation (143). Proposition 5.7 computes $b^{-\lambda}$ and $a^{-\lambda}$, while Proposition 5.8 computes $P_b(\beta)$ and $P_a(\beta)$.

Using Supplementary Equation (155), $|P_x(\beta)| \leq \frac{2x^{2l_3}}{|1 - \beta|}$, for $x \in \{a, b\}$, and because the constructions in Propositions 5.7 and 5.8 can be run with any fixed constant-factor tightening of the target precision without changing their polynomial resource scaling, we use the following working precisions. The endpoint-power oracle is run with pointwise error at most

$$\frac{\epsilon |1 - \beta|}{128 x^{2l_3}}, \quad (163)$$

and the polynomial oracle for $P_x(\beta)$ is run with error at most $\epsilon/64$. The propagated endpoint-power error then contributes at most $\epsilon/64$ to each product $x^{-\lambda} P_x(\beta)$, while the error in $P_x(\beta)$ contributes at most $\epsilon/64$ because $0 < x^{-\lambda} \leq 1$. The endpoint-power error is < 1 under the assumptions $0 < \epsilon < 1$ and $x > \beta + 2l_3$, so the product of the two input errors is also bounded by $\epsilon/64$. Hence each product in Supplementary Equation (162) is accurate to better than $\epsilon/16$ before the final arithmetic.

The final fixed-point multiplications and subtraction in Supplementary Equation (162) are performed with working precision chosen so that their total rounding and truncation contribution is at most $\epsilon/8$. Therefore $\widehat{S}(a, b, \beta)$ is an $\epsilon/4$ -approximation of $\widetilde{S}(a, b, \beta)$. By Proposition 5.5, $\widetilde{S}(a, b, \beta)$ is itself within $\epsilon/2$ of $S(a, b, \beta)$. Hence

$$|\widehat{S}(a, b, \beta) - S(a, b, \beta)| < \epsilon. \quad (164)$$

The stated resource bound follows by combining Propositions 5.7 and 5.8, together with the final fixed-point multiplication and subtraction in Supplementary Equation (162). $\square$

Extended state and fixed-point amplification

Corollary 5.10 (Extended state preparation). *Let $\beta > 0$, $\beta \neq 1$, and let $0 < \epsilon < 1$. Let $n_0 < n_1$ be integers satisfying the following conditions:*

1. $n_1 - n_0 = 2^k$ for some integer $k > 0$;
2. $n_0 - 1 = 2^{k_0}$ for some integer $k_0 \geq 0$, and $n_0 - 1$ is the smallest power of 2 larger than $\lceil \beta + 2c_{\text{ext}} \rceil$, where

$$\epsilon_S^{\text{ext}} = \frac{(\epsilon/3)^2 n_1^{-\beta}}{24k^2}, \quad c_{\text{ext}} = \left\lceil \frac{1}{2} \log_{2\pi} \left(\frac{8}{\epsilon_S^{\text{ext}}} \right) \right\rceil. \quad (165)$$

Define

$$|\psi_0\rangle_e = \frac{1}{C_e} \sum_{n=1}^{n_1-1} n^{-\beta/2} |n\rangle, \quad C_e = \sqrt{\sum_{n=1}^{n_1-1} n^{-\beta}}. \quad (166)$$

Then the state $|\psi_0\rangle_e$ can be prepared on a standard gate-based quantum computer to precision ϵ using

$$\text{Poly}\left(\log(1/\epsilon), \log(n_1), \log\left(\frac{1}{|1-\beta|}\right), \beta, v\right) \quad (167)$$

gates and ancilla qubits, where v denotes the number of bits of precision used to represent β .

Proof. We decompose the target state into its high-index and low-index parts. Define

$$C_T = \sqrt{\sum_{n=n_0}^{n_1-1} n^{-\beta}}, \quad C_L = \sqrt{\sum_{n=1}^{n_0-1} n^{-\beta}}, \quad C_e = \sqrt{C_T^2 + C_L^2}. \quad (168)$$

Let

$$|\psi_T\rangle = \frac{1}{C_T} \sum_{n=n_0}^{n_1-1} n^{-\beta/2} |n\rangle, \quad |\psi_L\rangle = \frac{1}{C_L} \sum_{n=1}^{n_0-1} n^{-\beta/2} |n\rangle. \quad (169)$$

These two states have disjoint supports and are therefore orthogonal. Moreover,

$$|\psi_0\rangle_e = \frac{C_T |\psi_T\rangle + C_L |\psi_L\rangle}{C_e} = \frac{|\psi_T\rangle + \mu |\psi_L\rangle}{\sqrt{1 + \mu^2}}, \quad \mu = \frac{C_L}{C_T}. \quad (170)$$

By Theorem 5.2, the truncated state $|\psi_T\rangle$ can be prepared to precision $\epsilon/3$ using

$$\text{Poly}\left(\log(1/\epsilon), \log(n_1), \log\left(\frac{1}{|1-\beta|}\right), \beta, v\right) \quad (171)$$

gates and ancilla qubits. The state constructed in Theorem 5.2 is initially encoded as $|n - n_0\rangle$. Adding n_0 to the computational register relabels the basis state as $|n\rangle$, with only $\mathcal{O}(\log n_1)$ additional gates.

For the low-index state $|\psi_L\rangle$, the support has size $n_0 - 1$, which is a power of 2 by assumption. Hence $|\psi_L\rangle$ can be prepared by direct recursive amplitude splitting over this finite support. The required partial sums are finite sums of length $n_0 - 1$, and can be evaluated classically to the required precision with cost polynomial in n_0 , $\log(1/\epsilon)$, β , and v . The resulting rotation angles are then classically precomputed and synthesized as controlled rotations to the same precision within this polynomial overhead. Therefore, $|\psi_L\rangle$ can be prepared to precision $\epsilon/3$ using

$$\text{Poly}(n_0, \log(1/\epsilon), \beta, v) \quad (172)$$

gates and ancilla qubits.

This cost is also polynomial in the parameters appearing in the statement. Since $n_0 - 1$ is chosen as the smallest power of 2 larger than $\lceil \beta + 2c_{\text{ext}} \rceil$, we have

$$n_0 \leq 2(\lceil \beta + 2c_{\text{ext}} \rceil + 1) + 1 = \mathcal{O}(\beta + c_{\text{ext}}). \quad (173)$$

Furthermore,

$$c_{\text{ext}} = \left\lceil \frac{1}{2} \log_{2\pi} \left(\frac{8}{\epsilon_S^{\text{ext}}} \right) \right\rceil, \quad \epsilon_S^{\text{ext}} = \frac{(\epsilon/3)^2 n_1^{-\beta}}{24k^2}. \quad (174)$$

Since $k = \log_2(n_1 - n_0) \leq \log_2 n_1$, it follows that

$$c_{\text{ext}} = \mathcal{O}(\log(1/\epsilon) + \beta \log n_1 + \log \log n_1). \quad (175)$$

Consequently,

$$n_0 = \text{Poly}(\beta, \log(n_1), \log(1/\epsilon)). \quad (176)$$

Thus the cost of preparing the low-index state is bounded by

$$\text{Poly}(\log(1/\epsilon), \log(n_1), \beta, v), \quad (177)$$

and is absorbed into the claimed overall complexity.

We combine the two orthogonal components using an LCU-type construction. Prepare an auxiliary qubit in the state

$$\frac{|0\rangle + \mu|1\rangle}{\sqrt{1 + \mu^2}}, \quad (178)$$

and, conditioned on this qubit, prepare $|\psi_T\rangle$ if the auxiliary qubit is $|0\rangle$, and $|\psi_L\rangle$ if it is $|1\rangle$. This gives

$$\frac{|0\rangle|\psi_T\rangle + \mu|1\rangle|\psi_L\rangle}{\sqrt{1 + \mu^2}}. \quad (179)$$

Applying a Hadamard gate to the auxiliary qubit yields

$$\frac{1}{\sqrt{2(1 + \mu^2)}} \left[|0\rangle(|\psi_T\rangle + \mu|\psi_L\rangle) + |1\rangle(|\psi_T\rangle - \mu|\psi_L\rangle) \right]. \quad (180)$$

Instead of measuring the auxiliary qubit, we perform the correction coherently. Conditioned on the auxiliary qubit being $|1\rangle$, apply a phase flip on the low-index subspace $1 \leq n < n_0$. Specifically, a comparator writes a flag indicating whether $n < n_0$, a controlled- Z gate flips the phase on this flagged subspace, and the comparator is then uncomputed:

$$\begin{aligned} |1\rangle(|\psi_T\rangle - \mu|\psi_L\rangle)|0\rangle &\xrightarrow{\text{flag}} |1\rangle(|\psi_T\rangle|0\rangle - \mu|\psi_L\rangle|1\rangle) \\ &\xrightarrow{Z} |1\rangle(|\psi_T\rangle|0\rangle + \mu|\psi_L\rangle|1\rangle) \\ &\xrightarrow{\text{unflag}} |1\rangle(|\psi_T\rangle + \mu|\psi_L\rangle)|0\rangle. \end{aligned} \quad (181)$$

The $|0\rangle$ branch is unchanged. Hence the joint state becomes

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|\psi_T\rangle + \mu|\psi_L\rangle}{\sqrt{1 + \mu^2}} = |+\rangle|\psi_0\rangle_e. \quad (182)$$

Applying a final Hadamard gate to the auxiliary qubit gives $|0\rangle|\psi_0\rangle_e$. Thus the combination step is a deterministic unitary procedure, up to the same arithmetic approximations used to prepare μ , $|\psi_T\rangle$, and $|\psi_L\rangle$.

For the approximation of μ , we have

$$\mu = \frac{C_L}{C_T} = \sqrt{\frac{S(1, n_0 - 1, \beta)}{S(n_0, n_1 - 1, \beta)}}. \quad (183)$$

The numerator $S(1, n_0 - 1, \beta)$ is a finite sum of length $\text{Poly}(\beta, \log n_1, \log(1/\epsilon))$, and can therefore be computed classically by direct summation. The denominator $S(n_0, n_1 - 1, \beta)$ is computed using Lemma 5.9. Since

$$S(n_0, n_1 - 1, \beta) \geq n_1^{-\beta}, \quad (184)$$

the denominator is bounded away from zero by an inverse exponential in $\beta \log n_1$. Therefore, computing the numerator and denominator to sufficiently high fixed-point precision within

$$\text{Poly}\left(\log(1/\epsilon), \log(n_1), \log\left(\frac{1}{1 - \beta}\right), \beta, v\right) \quad (185)$$

time suffices to obtain an estimate $\tilde{\mu}$ satisfying

$$|\tilde{\mu} - \mu| \leq \frac{\epsilon^2}{18}. \quad (186)$$

For orthogonal states $|\psi_T\rangle$ and $|\psi_L\rangle$, this implies

$$\left\| \frac{|\psi_T\rangle + \tilde{\mu}|\psi_L\rangle}{\sqrt{1 + \tilde{\mu}^2}} - \frac{|\psi_T\rangle + \mu|\psi_L\rangle}{\sqrt{1 + \mu^2}} \right\| = \sqrt{2 - 2 \frac{1 + \mu\tilde{\mu}}{\sqrt{(1 + \mu^2)(1 + \tilde{\mu}^2)}}} = \sqrt{2 - 2 \sqrt{1 - \frac{(\mu - \tilde{\mu})^2}{(1 + \mu^2)(1 + \tilde{\mu}^2)}}} < \sqrt{2|\mu - \tilde{\mu}|} \leq \frac{\varepsilon}{3}. \quad (187)$$

Let $|\tilde{\psi}_T\rangle$ and $|\tilde{\psi}_L\rangle$ be the prepared high- and low-index states, each within distance $\varepsilon/3$ of its target, and let $\tilde{\mu}$ be the computed coefficient. Before the Hadamard and coherent correction, the implemented joint state is

$$|\tilde{\Phi}\rangle = \frac{|0\rangle|\tilde{\psi}_T\rangle + \tilde{\mu}|1\rangle|\tilde{\psi}_L\rangle}{\sqrt{1 + \tilde{\mu}^2}}. \quad (188)$$

By orthogonality of the auxiliary branches,

$$\left\| |\tilde{\Phi}\rangle - \frac{|0\rangle|\psi_T\rangle + \tilde{\mu}|1\rangle|\psi_L\rangle}{\sqrt{1 + \tilde{\mu}^2}} \right\| \leq \frac{\varepsilon}{3}. \quad (189)$$

The coefficient estimate above gives

$$\left\| \frac{|0\rangle|\psi_T\rangle + \tilde{\mu}|1\rangle|\psi_L\rangle}{\sqrt{1 + \tilde{\mu}^2}} - \frac{|0\rangle|\psi_T\rangle + \mu|1\rangle|\psi_L\rangle}{\sqrt{1 + \mu^2}} \right\| \leq \frac{\varepsilon}{3}. \quad (190)$$

Hence the state entering the coherent combination is within $2\varepsilon/3$ of the ideal one. The subsequent Hadamard gates, controlled low-index phase flip, and comparator uncomputation are unitary, so they preserve this distance. Therefore the final joint state is within $2\varepsilon/3 < \varepsilon$ of $|0\rangle|\psi_0\rangle_e$. The total number of gates and ancilla qubits is bounded by

$$\text{Poly}\left(\log(1/\varepsilon), \log(n_1), \log\left(\frac{1}{|1 - \beta|}\right), \beta, v\right), \quad (191)$$

as claimed. $\square$

Theorem 5.11 (Initial state preparation (Methods, Theorem 3)). *Let $\beta > 0$, $\beta \neq 1$, and $N \in \mathbb{N}$. Define*

$$|\psi_0\rangle = \frac{1}{C_N} \sum_{n=1}^N n^{-\beta/2} |n\rangle, \quad C_N = \sqrt{\sum_{n=1}^N n^{-\beta}}. \quad (192)$$

Then $|\psi_0\rangle$ can be prepared deterministically on a quantum computer to precision $\varepsilon > 0$. The required number of gates and ancilla qubits is bounded by

$$\text{Poly}(\log \varepsilon^{-1}, \log N, \log |1 - \beta|^{-1}, \beta, v), \quad (193)$$

where v denotes the number of bits of precision used to represent β .

Proof. Set

$$\varepsilon_{\text{ps}} = \frac{\varepsilon}{4}, \quad \delta = \frac{\varepsilon_{\text{ps}}}{6}. \quad (194)$$

We first construct a constant-success coherent preparation primitive, and then amplify its successful component to make the final output deterministic up to error ε .

Choose the lower cutoff n_0 large enough for the Euler–Maclaurin-based partial-sum oracle used in the extended-state preparation. When Corollary 5.10 is applied with target precision δ , its partial-sum precision parameter is

$$\epsilon_S^{\text{ext}} = \frac{(\delta/3)^2 n_1^{-\beta}}{24k^2} = \frac{\varepsilon_{\text{ps}}^2 n_1^{-\beta}}{7776k^2}. \quad (195)$$

Since the construction below will satisfy

$$n_1 \leq 2N, \quad k = \log(n_1 - n_0) \leq \lceil \log(2N) \rceil, \quad (196)$$

it is sufficient to define

$$c_{\text{init}} = \left\lceil \frac{1}{2} \log_{2\pi} \left(\frac{62208 \lceil \log(2N) \rceil^2 (2N)^\beta}{\varepsilon_{\text{ps}}^2} \right) \right\rceil. \quad (197)$$

Choose n_0 such that $n_0 - 1$ is the smallest power of 2 larger than $\lceil \beta + 2c_{\text{init}} \rceil$. Then

$$n_0 = \text{Poly}(\beta, \log N, \log(1/\varepsilon)). \quad (198)$$

If $N \leq n_0$, then the desired state is supported on only $\text{Poly}(\beta, \log N, \log(1/\varepsilon))$ basis states and can be prepared directly by recursive amplitude splitting within the claimed resource bound. Hence, in the following, assume $N > n_0$.

Let n_1 be the smallest integer larger than N such that $n_1 - n_0$ is a power of 2, and define

$$k = \log(n_1 - n_0). \quad (199)$$

Since $N > n_0$, this choice implies $n_1 < 2N + 1$, and hence $n_1 \leq 2N$ and $k \leq \lceil \log(2N) \rceil$. By Supplementary Equation (197), the assumptions of Corollary 5.10 are satisfied.

By Corollary 5.10, the extended state

$$|\psi_e\rangle = \frac{1}{C_e} \sum_{n=1}^{n_1-1} n^{-\beta/2} |n\rangle, \quad C_e = \sqrt{\sum_{n=1}^{n_1-1} n^{-\beta}}, \quad (200)$$

can be prepared to precision δ using

$$\text{Poly}\left(\log(1/\varepsilon), \log(N), \log\left(\frac{1}{|1-\beta|}\right), \beta, v\right) \quad (201)$$

gates and ancilla qubits.

Let

$$P_N = \sum_{n=1}^N |n\rangle\langle n| \quad (202)$$

be the projector onto the target subspace. Applying P_N to the ideal extended state gives

$$\frac{P_N |\psi_e\rangle}{\sqrt{\langle \psi_e | P_N | \psi_e \rangle}} = \frac{1}{C_N} \sum_{n=1}^N n^{-\beta/2} |n\rangle = |\psi_0\rangle. \quad (203)$$

The ideal success probability is

$$p = \langle \psi_e | P_N | \psi_e \rangle = \frac{\sum_{n=1}^N n^{-\beta}}{\sum_{n=1}^{n_1-1} n^{-\beta}}. \quad (204)$$

Since $n^{-\beta}$ is positive and decreasing for $\beta > 0$, we have

$$p \geq \frac{N}{n_1 - 1}. \quad (205)$$

Moreover, by the choice of n_1 ,

$$n_1 - 1 \leq 2(N - n_0) + n_0 - 1 = 2N - n_0 - 1 \leq 2N - 3. \quad (206)$$

Therefore,

$$p \geq \frac{N}{2N - 3} > \frac{1}{2}. \quad (207)$$

Now let $|\tilde{\psi}_e\rangle$ denote the actually prepared extended state, satisfying

$$\left\| |\tilde{\psi}_e\rangle - |\psi_e\rangle \right\| \leq \delta. \quad (208)$$

Let

$$\tilde{p} = \langle \tilde{\psi}_e | P_N | \tilde{\psi}_e \rangle \quad (209)$$

be the corresponding success probability, and define the normalized successful branch

$$|\tilde{\psi}_0\rangle = \frac{P_N |\tilde{\psi}_e\rangle}{\sqrt{\tilde{p}}}. \quad (210)$$

Since $0 \leq P_N \leq I$,

$$\begin{aligned} |p - \tilde{p}| &= \left| \langle \psi_e | P_N | \psi_e \rangle - \langle \tilde{\psi}_e | P_N | \tilde{\psi}_e \rangle \right| \\ &\leq \left| \langle \psi_e | P_N (|\psi_e\rangle - |\tilde{\psi}_e\rangle) \right| + \left| (\langle \psi_e | - \langle \tilde{\psi}_e |) P_N |\tilde{\psi}_e\rangle \right| \\ &\leq 2\delta = \frac{\varepsilon_{\text{ps}}}{3}. \end{aligned} \quad (211)$$

Thus

$$\tilde{p} \geq p - \frac{\varepsilon_{\text{ps}}}{3} > \frac{1}{2} - \frac{\varepsilon_{\text{ps}}}{3} > \frac{1}{3}. \quad (212)$$

Furthermore, by contractivity of P_N ,

$$\left\| P_N |\psi_e\rangle - P_N |\tilde{\psi}_e\rangle \right\| \leq \delta. \quad (213)$$

Equivalently,

$$\left\| \sqrt{p} |\psi_0\rangle - \sqrt{\tilde{p}} |\tilde{\psi}_0\rangle \right\| \leq \delta. \quad (214)$$

Also,

$$|\sqrt{p} - \sqrt{\tilde{p}}| = \frac{|p - \tilde{p}|}{\sqrt{p} + \sqrt{\tilde{p}}} \leq \sqrt{2} |p - \tilde{p}| \leq 2\sqrt{2} \delta, \quad (215)$$

where we used $p \geq 1/2$. Therefore,

$$\begin{aligned} \sqrt{p} \left\| |\psi_0\rangle - |\tilde{\psi}_0\rangle \right\| &\leq \left\| \sqrt{p} |\psi_0\rangle - \sqrt{\tilde{p}} |\tilde{\psi}_0\rangle \right\| + |\sqrt{p} - \sqrt{\tilde{p}}| \\ &\leq \delta + 2\sqrt{2} \delta. \end{aligned} \quad (216)$$

Since $p \geq 1/2$, we obtain

$$\left\| |\psi_0\rangle - |\tilde{\psi}_0\rangle \right\| \leq \sqrt{2}(1 + 2\sqrt{2})\delta = \frac{1 + 2\sqrt{2}}{3\sqrt{2}} \varepsilon_{\text{ps}} < \varepsilon_{\text{ps}}. \quad (217)$$

It remains to remove the post-selection. The projection onto the target subspace can be implemented coherently by a comparator that writes a success flag:

$$|n\rangle|0\rangle_f \mapsto |n\rangle|f_N(n)\rangle_f, \quad f_N(n) = \begin{cases} 1, & 1 \leq n \leq N, \\ 0, & N < n \leq n_1 - 1. \end{cases} \quad (218)$$

Let A denote the coherent circuit consisting of the extended-state preparation followed by this comparator. Then

$$A|0\rangle = \sqrt{\tilde{p}} |\tilde{\psi}_0\rangle |1\rangle_f + \sqrt{1 - \tilde{p}} |\tilde{\psi}_\perp\rangle |0\rangle_f, \quad (219)$$

for some normalized state $|\tilde{\psi}_\perp\rangle$ supported on the complement of the target subspace. The good subspace is the flag-1 subspace, and its initial weight satisfies $\tilde{p} > 1/3$.

We now apply fixed-point amplitude amplification³ to the coherent primitive A , using the known lower bound $p_0 = 1/3$ on the good probability. Choose the target failure probability

$$\epsilon_{\text{amp}} = \frac{\varepsilon^2}{64}. \quad (220)$$

Fixed-point amplitude amplification produces a unitary circuit $\mathcal{A}_N$ such that

$$\mathcal{A}_N |0\rangle = \sqrt{1 - q} |\tilde{\psi}_0\rangle |1\rangle_f + \sqrt{q} |\text{bad}\rangle, \quad 0 \leq q \leq \epsilon_{\text{amp}}, \quad (221)$$

where $|\text{bad}\rangle$ is normalized and lies in the flag-0 subspace. Hence

$$\left\| \mathcal{A}_N |0\rangle - |\tilde{\psi}_0\rangle |1\rangle_f \right\| \leq \left| 1 - \sqrt{1 - q} \right| + \sqrt{q} \leq 2\sqrt{q} \leq 2\sqrt{\epsilon_{\text{amp}}} = \frac{\varepsilon}{4}. \quad (222)$$

Combining this with $\| |\tilde{\psi}_0\rangle - |\psi_0\rangle \| < \varepsilon_{\text{ps}} = \varepsilon/4$, we obtain

$$d(\mathcal{A}_N |0\rangle, |\psi_0\rangle |1\rangle_f |0\rangle_{\text{work}}) \leq \frac{\varepsilon}{4} + \frac{\varepsilon}{4} < \varepsilon, \quad (223)$$

where any additional fixed-point phases, reflections, and single-qubit rotations are synthesized to sufficiently high precision and absorbed into the same ε budget.

Finally, because the initial good probability is lower bounded by the constant $p_0 = 1/3$, the fixed-point amplification requires only $\mathcal{O}(\log(1/\epsilon_{\text{amp}})) = \mathcal{O}(\log(1/\varepsilon))$ coherent uses of A , $A^\dagger$, and the corresponding good-state and initial-state reflections. These operations increase the gate count only by a polynomial factor in $\log(1/\varepsilon)$. They are applied sequentially, so the number of ancilla qubits remains polynomial in

$$\log(1/\varepsilon), \quad \log(N), \quad \log\left(\frac{1}{1 - \beta}\right), \quad \beta, \quad v. \quad (224)$$

Thus $|\psi_0\rangle$ is prepared deterministically to precision ε , with the stated resource scaling. $\square$

Supplementary Note 6 – Construction of the evolution operator

This note proves that the evolution operator generated by the logarithmic Hamiltonian $\mathcal{H}_0$ can be implemented with polynomial quantum resources.

Theorem 6.1 (Evolution operator construction (Methods, Theorem 4)). *Define the time evolution operator*

$$U(t) = e^{-i\mathcal{H}_0 t}, \quad \mathcal{H}_0 = \sum_{n=1}^N \ln(n) |n\rangle\langle n|. \quad (225)$$

Suppose that the evolution time t , with $|t| \geq 2$, is specified with u bits of precision. Then, for any $0 < \xi < 1$, $U(t)$ can be implemented to precision ξ using

$$\text{Poly}(\log(N), \log |t|, \log(1/\xi), u) \quad (226)$$

gates and ancilla qubits.

Proof. Consider an arbitrary input state

$$\sum_{n=1}^N \alpha_n |n\rangle |0\rangle, \quad (227)$$

where $|0\rangle$ denotes the ancilla register. By Lemma 4.3, we can coherently compute the base-2 logarithm of n :

$$\sum_{n=1}^N \alpha_n |n\rangle |0\rangle \mapsto \sum_{n=1}^N \alpha_n |n\rangle |\widetilde{\log(n)}\rangle. \quad (228)$$

Since $\ln(n) = (\ln 2) \log(n)$, we multiply the logarithm register by $(\ln 2)t$ to obtain an approximation of the phase $\ln(n)t$:

$$\sum_{n=1}^N \alpha_n |n\rangle |\widetilde{\log(n)}\rangle \mapsto \sum_{n=1}^N \alpha_n |n\rangle |\widetilde{\phi_n}\rangle, \quad \widetilde{\phi_n} \approx \ln(n)t. \quad (229)$$

Applying controlled phase rotations conditioned on the bits of the phase register gives

$$\sum_{n=1}^N \alpha_n e^{-i\widetilde{\phi_n}} |n\rangle |\widetilde{\phi_n}\rangle. \quad (230)$$

Finally, we uncompute the multiplication and uncompute the logarithm register, leaving

$$\sum_{n=1}^N \alpha_n e^{-i\widetilde{\phi_n}} |n\rangle |0\rangle. \quad (231)$$

In the ideal case, $\widetilde{\phi_n} = \ln(n)t$, and hence the resulting operation is exactly

$$\sum_{n=1}^N \alpha_n e^{-i\ln(n)t} |n\rangle = U(t) \sum_{n=1}^N \alpha_n |n\rangle. \quad (232)$$

For the precision, compute $\log(n)$ to precision $\eta_{\log} = \frac{\xi}{6|t|\ln 2}$. Then the induced error in $(\ln 2)t \log(n) = t \ln(n)$ is at most $|t| \ln 2 \eta_{\log} \leq \frac{\xi}{6}$. The multiplication by $(\ln 2)t$ and the fixed-point representation of the phase register are performed with sufficient precision so that their additional phase error is at most $\xi/6$. The controlled phase rotation network is synthesized so that its total contribution to the implemented unitary error is at most $\xi/6$. Therefore, for every n , $|\widetilde{\phi_n} - \ln(n)t| \leq \frac{\xi}{2}$. Consequently, for any normalized input state,

$$\left\| \sum_{n=1}^N \alpha_n \left(e^{-i\widetilde{\phi_n}} - e^{-i\ln(n)t} \right) |n\rangle \right\| \leq \max_{1 \leq n \leq N} \left| e^{-i\widetilde{\phi_n}} - e^{-i\ln(n)t} \right| \leq 2 \sin\left(\frac{1}{2} \max_n |\widetilde{\phi_n} - \ln(n)t|\right) \leq \xi. \quad (233)$$

Thus, the implemented unitary is within distance ξ of $U(t)$.

For the resource estimate, Lemma 4.3 implies that computing $\log(n)$ to precision $\eta_{\log}$ requires

$$\mathcal{O}\left((\log N)^3 \log^2\left(\frac{|t|}{\xi}\right) + (\log N)^2 \log^3\left(\frac{|t|}{\xi}\right)\right) \quad (234)$$

gates and

$$\mathcal{O}\left(\log N \log^2\left(\frac{|t|}{\xi}\right)\right) \quad (235)$$

ancilla qubits, up to constant factors. The uncomputation of the logarithm oracle has the same cost.

The phase $\ln(n)t$ has magnitude at most $|t| \ln(N)$. Thus the phase register requires

$$\mathcal{O}(\log |t| + \log \log(N) + \log(1/\xi)) \quad (236)$$

integer and fractional bits. Multiplication by the u -bit number t , including the constant factor $\ln 2$, can therefore be performed with gate and ancilla complexity polynomial in $\log(N)$, $\log |t|$, $\log(1/\xi)$, and u . The controlled phase rotations are implemented bit by bit from the phase register and require a number of gates polynomial in the phase-register length and $\log(1/\xi)$. Combining the logarithm oracle, arithmetic multiplication, controlled phase rotations, and uncomputation, the total resource requirement is

$$\text{Poly}(\log(N), \log |t|, \log(1/\xi), u) \quad (237)$$

gates and ancilla qubits. This completes the proof. $\square$

Supplementary Note 7 – Order of $|\chi(s)|$ for $0 < \beta < 1$

This note estimates the growth of the functional-equation factor $\chi(s)$ in the critical strip.

Recall that

$$\chi(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s). \quad (238)$$

Thus, for $s = \beta + it$ with $0 < \beta < 1$,

$$|\chi(s)| = 2^\beta \pi^{\beta-1} \left| \sin\left(\frac{\pi s}{2}\right) \right| |\Gamma(1-s)|. \quad (239)$$

We first note that the elementary prefactor is uniformly bounded in β :

$$\frac{1}{\pi} \leq 2^\beta \pi^{\beta-1} \leq 2, \quad 0 < \beta < 1. \quad (240)$$

For the sine factor, since

$$\sin\left(\frac{\pi s}{2}\right) = \sin\left(\frac{\pi \beta}{2}\right) \cosh\left(\frac{\pi t}{2}\right) + i \cos\left(\frac{\pi \beta}{2}\right) \sinh\left(\frac{\pi t}{2}\right), \quad (241)$$

we have

$$\begin{aligned} \left| \sin\left(\frac{\pi s}{2}\right) \right| &= \sqrt{\sin^2\left(\frac{\pi \beta}{2}\right) \cosh^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi \beta}{2}\right) \sinh^2\left(\frac{\pi t}{2}\right)} \\ &= \sqrt{\sinh^2\left(\frac{\pi t}{2}\right) + \sin^2\left(\frac{\pi \beta}{2}\right)}. \end{aligned} \quad (242)$$

Therefore, uniformly for $0 < \beta < 1$ as $|t| \rightarrow \infty$,

$$\left| \sin\left(\frac{\pi(\beta + it)}{2}\right) \right| = \Theta\left(e^{\pi|t|/2}\right), \quad |t| \rightarrow \infty. \quad (243)$$

Next, we estimate the Gamma factor. By Stirling's formula, uniformly in vertical strips of bounded real part,

$$|\Gamma(\sigma + i\tau)| = \Theta\left(|\tau|^{\sigma - \frac{1}{2}} e^{-\pi|\tau|/2}\right), \quad |\tau| \rightarrow \infty, \quad (244)$$

where the implicit constants are uniform for $0 \leq \sigma \leq 1$. Taking

$$\sigma = 1 - \beta, \quad \tau = -t, \quad (245)$$

we obtain, uniformly for $0 < \beta < 1$ as $|t| \rightarrow \infty$,

$$|\Gamma(1 - \beta - it)| = \Theta\left(|t|^{\frac{1}{2} - \beta} e^{-\pi|t|/2}\right), \quad |t| \rightarrow \infty. \quad (246)$$

Combining the prefactor bound, the sine estimate, and the Gamma estimate, the exponential factors cancel. Hence there exist absolute constants $K_-, K_+ > 0$ and $T > 0$, independent of β , such that for all $0 < \beta < 1$ and all $|t| \geq T$,

$$K_- |t|^{\frac{1}{2} - \beta} \leq |\chi(\beta + it)| \leq K_+ |t|^{\frac{1}{2} - \beta}. \quad (247)$$

Equivalently,

$$|\chi(\beta + it)| = \Theta\left(|t|^{\frac{1}{2} - \beta}\right), \quad |t| \rightarrow \infty, \quad (248)$$

uniformly for $0 < \beta < 1$, for all sufficiently large $|t|$.

Supplementary Note 8 – Upper bound of $\zeta'(s)$ for $0 < \beta < 1$

This note gives a polynomial upper bound on $\zeta'(s)$ in the critical strip using the eta-function representation.

Let

$$\eta(s) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s} \quad (249)$$

be the alternating Dirichlet series. For $\operatorname{Re}(s) = \beta > 0$, define the partial-sum function

$$\mathcal{A}(x) = \sum_{1 \leq n \leq \lfloor x \rfloor} (-1)^{n+1}. \quad (250)$$

Then $|\mathcal{A}(x)| \leq 1$ for all $x \geq 1$. Abel summation gives, for every $N \geq 1$,

$$\sum_{n=1}^N \frac{(-1)^{n+1}}{n^s} = \mathcal{A}(N)N^{-s} + s \int_1^N \mathcal{A}(x)x^{-s-1} dx. \quad (251)$$

Since $|\mathcal{A}(N)N^{-s}| \leq N^{-\beta} \rightarrow 0$, letting $N \rightarrow \infty$ yields

$$\eta(s) = s \int_1^{\infty} \mathcal{A}(x)x^{-s-1} dx. \quad (252)$$

Since $|\mathcal{A}(x)| \leq 1$, we obtain

$$|\eta(s)| \leq |s| \int_1^{\infty} x^{-\beta-1} dx = \frac{|s|}{\beta}. \quad (253)$$

The Abel-summation representation may be differentiated under the integral sign because, on any compact substrip $\operatorname{Re}(s) \geq \beta_0 > 0$, the derivative is dominated by an integrable multiple of $(1 + \ln x)x^{-\beta_0-1}$. Differentiating gives

$$\eta'(s) = \int_1^{\infty} \mathcal{A}(x)x^{-s-1} dx - s \int_1^{\infty} \mathcal{A}(x) \ln(x)x^{-s-1} dx. \quad (254)$$

Hence

$$|\eta'(s)| \leq \int_1^{\infty} x^{-\beta-1} dx + |s| \int_1^{\infty} \ln(x)x^{-\beta-1} dx = \frac{1}{\beta} + \frac{|s|}{\beta^2}. \quad (255)$$

The zeta function is related to the eta function by $\eta(s) = (1 - 2^{1-s}) \zeta(s)$, and hence $\zeta(s) = \frac{\eta(s)}{1-2^{1-s}}$. Differentiating gives

$$\zeta'(s) = \frac{\eta'(s)}{1 - 2^{1-s}} - \frac{2^{1-s} \ln 2 \eta(s)}{(1 - 2^{1-s})^2}. \quad (256)$$

Therefore,

$$|\zeta'(s)| \leq \frac{|s|/\beta^2 + 1/\beta}{|1 - 2^{1-s}|} + \frac{2^{1-\beta} \ln 2 |s|/\beta}{|1 - 2^{1-s}|^2}. \quad (257)$$

For $0 < \beta < 1$, we have

$$|1 - 2^{1-s}| = \left| 1 - 2^{1-\beta} e^{-it \ln 2} \right| \geq 2^{1-\beta} - 1 \geq (\ln 2)(1 - \beta), \quad (258)$$

where the last inequality follows from $e^x - 1 \geq x$ for $x \geq 0$. Substituting this lower bound yields

$$|\zeta'(s)| = \mathcal{O}\left(\frac{|s| + 1}{\beta^2 |1 - \beta|} + \frac{|s|}{\beta |1 - \beta|^2}\right). \quad (259)$$

Since $0 < \beta < 1$, we have $|s| \leq |t| + 1$. Hence

$$|\zeta'(s)| = \operatorname{Poly}\left(|t|, \frac{1}{\beta}, \frac{1}{|1 - \beta|}\right). \quad (260)$$

Supplementary References

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